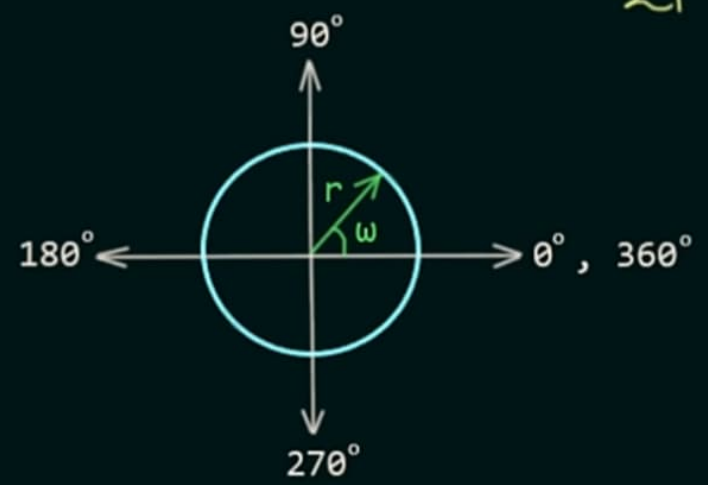


12T

$n = -\infty \leq$

Bidirectional
ZT

Complex variable
 $Z = re^{j\omega}$
 $r = |Z|$ (radius)
 ω (complex arg.)



Ex. :- $x[n] = a^n u[n] \Rightarrow X(z) = ?$
 ROC = ?

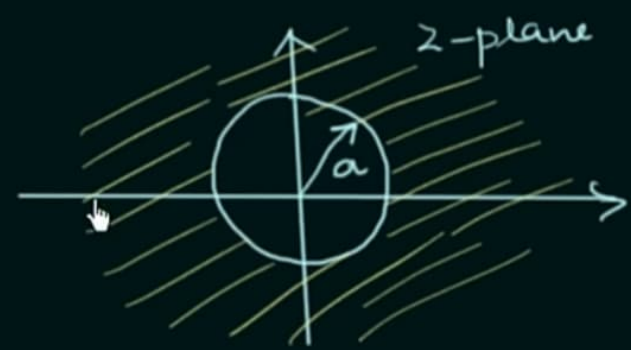
Sol. :- $X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] \cdot z^{-n}$

$= \sum_{n=0}^{\infty} a^n z^{-n}$

$S_{\infty} = \frac{F.T.}{1 - C.R.}$, $|C.R.| < 1$

$= \sum_{n=0}^{\infty} (az^{-1})^n$ (C.R. = az^{-1})
 $= 1 + az^{-1} + (az^{-1})^2 + \dots$

$X(z) = \frac{1}{1 - az^{-1}}$, $|az^{-1}| < 1$
 $|\frac{a}{z}| < 1$



$r = |Z| > a$
ROC

Introduction to Z-Transform

$$\begin{array}{c} \text{E\&P} \\ \uparrow \\ \text{CTFT} \end{array} \equiv \begin{array}{c} \text{E\&P} \\ \uparrow \\ \text{DTFT} \end{array}$$

$$\begin{array}{c} \text{LT} \\ \downarrow \\ \text{E, P, \& NENP}^* \end{array} \equiv \text{ZT} \rightarrow \text{E, P, \& NENP}^*$$

$$\rightarrow X[z] = \sum_{n=0}^{\infty} x[n] \cdot z^{-n}$$

Unidirectional
ZT

$$\rightarrow x[n] \xrightleftharpoons[\text{IZT}]{\text{ZT}} X[z] = \sum_{n=-\infty}^{\infty} x[n] \cdot \underbrace{z^{-n}}_{\substack{\text{int.} \\ \text{complex} \\ \text{variable}}} \xrightarrow{\text{Bidirectional ZT}}$$

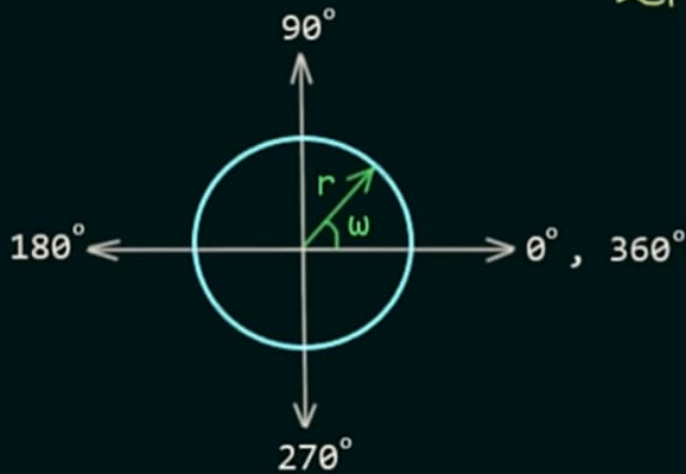
$$\text{Ex. :- } x[n] = a^n u[n] \Rightarrow X[z] = ? \\ \text{Roc} = ?$$

$$\text{Sol. :- } X[z] = \sum_{n=-\infty}^{\infty} a^n u[n] \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (a z^{-1})^n$$

$$= 1$$



Condition for Existence of Z-Transform

$$x[n] \Rightarrow X[z] = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}$$

$z = re^{j\omega}$

$$\rightarrow |X[z]| < \infty$$

$$\left| \sum_{n=-\infty}^{\infty} x[n] z^{-n} \right| < \infty$$

$$\sum_{n=-\infty}^{\infty} |x[n] z^{-n}| < \infty$$

$$\sum_{n=-\infty}^{\infty} |x[n] \cdot (re^{j\omega})^{-n}| < \infty$$

$$\sum_{n=-\infty}^{\infty} |x[n] r^{-n}| e^{-j\omega n} < \infty$$

$$\sum_{n=-\infty}^{\infty} |x[n] \tilde{r}^{-n}| < \infty$$

$r=1$

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

DTFT

$$\text{Ex: } x[n] = z^n u[n], \text{ ROC} = ?$$

$$\rightarrow \sum_{n=-\infty}^{\infty} |z^n u[n] \tilde{r}^{-n}| = \sum_{n=0}^{\infty} |z^n \tilde{r}^{-n}|$$

$$= \sum_{n=0}^{\infty} |(z \tilde{r}^{-1})^n|$$

$$= \sum_{n=0}^{\infty} (z \tilde{r}^{-1})^n$$

$$= 1 + z \tilde{r}^{-1} + (z \tilde{r}^{-1})^2 + \dots$$

$$\Rightarrow 1 + z \tilde{r}^{-1} + (z \tilde{r}^{-1})^2 + \dots < \infty$$

Z transform by Neso Acade... • 2/10

Vedpriya Gautam



Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Unilateral Z-Transform

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

Bilateral Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Complex Plane

12:35

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Introduction to Z-Transform

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Condition for Existence of Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Condition for Existence of Z-Transform

$$\sum_{n=-\infty}^{\infty} |x[n] z^{-n}| < \infty$$

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Condition for Existence of ...

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Z-Transform Properties PART 1

$$1) \text{ Linearity: } X(z) \Rightarrow x[n] \cdot a + x[n] \cdot b \Rightarrow aX(z) + bX(z)$$

$$2) \text{ Time Shifting: } X(z) \Rightarrow x[n-k] \Rightarrow z^{-k} X(z)$$

$$3) \text{ Time Scaling: } X(z) \Rightarrow x[kn] \Rightarrow \frac{1}{k} X(z^k)$$

$$4) \text{ Time Reversal: } X(z) \Rightarrow x[-n] \Rightarrow X(1/z)$$

$$5) \text{ Scaling of Z: } X(z) \Rightarrow X(kz)$$

8:39

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Properties of Z-Transform (Part 1)

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Z-Transform Properties PART 2

$$1) \text{ Time Shifting: } X(z) \Rightarrow x[n-k] \Rightarrow z^{-k} X(z)$$

$$2) \text{ Time Scaling: } X(z) \Rightarrow x[kn] \Rightarrow \frac{1}{k} X(z^k)$$

$$3) \text{ Time Reversal: } X(z) \Rightarrow x[-n] \Rightarrow X(1/z)$$

$$4) \text{ Scaling of Z: } X(z) \Rightarrow X(kz)$$

8:41

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Properties of Z-Transform (Part 2)

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Z-Transform Properties PART 3

$$1) \text{ Conjugation: } X(z) \Rightarrow x^*[n] \Rightarrow X^*(1/z^*)$$

$$2) \text{ Accumulation: } X(z) \Rightarrow \sum_{k=-\infty}^n x[k] \Rightarrow \frac{X(z)}{1-z^{-1}}$$

9:31

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Properties of Z-Transform (Part 3)

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Z-Transform PART 4

Properties of Z-Transform

Properties of Z-Transform (Part 1)

1) Linearity:

$$x_1[n] \Rightarrow X_1[z], \text{ ROC} = R_1$$

$$x_2[n] \Rightarrow X_2[z], \text{ ROC} = R_2$$

$$\alpha x_1[n] \Rightarrow \alpha X_1[z], \text{ ROC} = R_1$$

$$\beta x_2[n] \Rightarrow \beta X_2[z], \text{ ROC} = R_2$$

$$\alpha x_1[n] + \beta x_2[n] \Rightarrow \alpha X_1[z] + \beta X_2[z], \\ \text{ROC} \geq R_1 \cap R_2$$

Proof:- H.W.

2) Time Shifting:

$$x[n] \Rightarrow X[z], \text{ ROC} = R$$

$$x[n-n_0] \Rightarrow z^{-n_0} X[z], \text{ ROC} = \text{same}$$

$$\text{Proof:- } x[n] \Rightarrow X[z] = \sum_{n=-\infty}^{\infty} x[n] z^n$$

$$x[n-n_0] \Rightarrow X'[z] = \sum_{n=-\infty}^{\infty} x[n-n_0] z^n$$

$$n-n_0 = m \Rightarrow n = n_0 + m$$

$$X'[z] = \sum_{m=-\infty}^{\infty} x[m] z^{-(n_0+m)}$$

$$X'[z] = z^{-n_0} \sum_{m=-\infty}^{\infty} x[m] z^{-m} \Rightarrow X'[z] = z^{-n_0} X[z]$$

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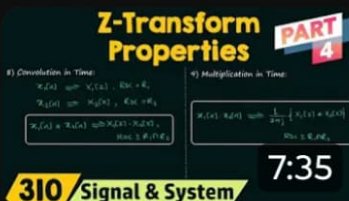
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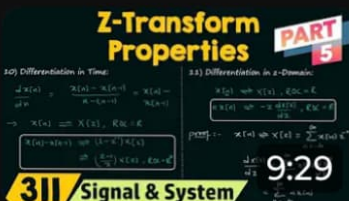
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Properties of Z-Transform (Part 4)

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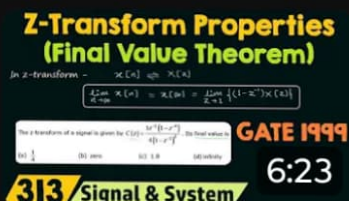
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Properties of Z-Transform (Part 5)

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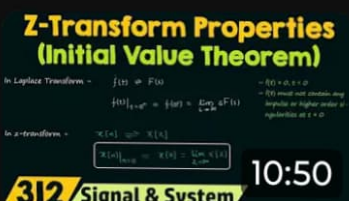
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Properties of Z-Transform (Final Value Theorem)

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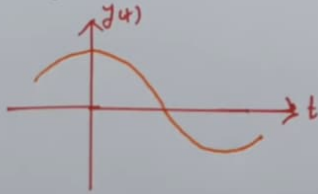
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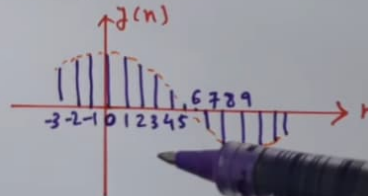
Continuous

- Time is Continuous
- Signal is defined for all values of time t .
- Independent variable is time t .
- graphical representation



Discrete

- Time is discrete
- Signal is defined for discrete interval of time t .
- Independent variable is sample, denoted by n .
- graphical representation



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- $f(t) \rightarrow f(s) \rightarrow$ Fourier transform
 $f(t) \rightarrow f(s) \rightarrow$ Laplace transform
- Z - Transform
- Analysis of discrete time LTI systems can be done by using Z - transforms
- Continuous
- Time is Continuous
 - Signal is defined for all values of time t .
 - Independent variable is time t .
 - graphical representation
- Discrete
- Time is discrete
 - Signal is defined for discrete values of time t .
 - Independent variable is discrete time n .
 - graphical representation

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Z-transform and Inverse Z-transform

- The Z-transform of a discrete time signal $x(n)$ is represented by $X(z)$

$$x(n) \xrightarrow{ZT} X(z)$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

- $x(t) \rightarrow \text{freq} \rightarrow \text{F.T.}, \text{L.T.}$
 $x(n) \rightarrow \text{freq} \rightarrow \text{D.F.T.}, \text{Z.T.}$

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for standard basic signal

7:52

1. Z Transform Examples: Standard Basic Signal in Sig...

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2. Z Transform Examples:

- Relation bet. DFT & Z Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$z = re^{j\omega}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) (re^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) r^{-n} e^{-j\omega n}$$

$$= \text{DFT}[x(n)r^n]$$

$$\text{DFT}[x(n)] = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

- Inverse Z - transform

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$$\begin{aligned}
 - x(z) &= \sum_{n=-\infty}^{\infty} x(n) (\gamma e^{j\omega})^{-n} \\
 &= \sum_{n=-\infty}^{\infty} x(n) \gamma^{-n} e^{-j\omega n} \\
 &= \text{DFT} [x(n) \gamma^{-n}]
 \end{aligned}$$

$$\text{DFT} [x(n)] = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

- Inverse Z - transform

$$\Rightarrow x(z) = \text{DFT} [x(n) \gamma^{-n}]$$

$$\Rightarrow \text{IDFT} [x(z)] = x(n) \gamma^{-n}$$

$$\Rightarrow x(n) = \gamma^n [\text{IDFT} (x(z))]$$

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$$= \sum_{n=-\infty}^{\infty} \frac{x(n) z^{-n} e^{-j\omega n}}{z^{-n} e^{-j\omega n}}$$

$$= \text{DFT} [x(n) z^{-n}]$$

Inverse Z - transform

$$\Rightarrow x(z) = \text{DFT} [x(n) z^{-n}]$$

$$\Rightarrow \text{IDFT} [x(z)] = x(n) z^{-n}$$

$$\Rightarrow x(n) = z^n [\text{IDFT} (x(z))] \\ = z^n \left[\frac{1}{2\pi} \int x(z) e^{j\omega n} d\omega \right]$$

$$= \frac{1}{2\pi} \int x(z) (z e^{j\omega})^n d\omega$$

$$\rightarrow z = z e^{j\omega}$$

$$\Rightarrow z = z e^{j\omega}$$

$$\Rightarrow dz = z e^{j\omega} j d\omega$$

$$\Rightarrow dz = jz d\omega$$

$$\Rightarrow d\omega = \frac{dz}{jz}$$

$$\rightarrow x(n) = \frac{1}{2\pi} \int x(z)$$

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2. Z Transform Examples:

Z-transform for finite sequence

A finite sequence $x(n)$ is defined as $x(n) = \{5, 3, -3, 0, 4, -2\}$
find $X(z)$ of given sequence.

$$- x(n) = \{ \overset{x(-2)}{5}, \overset{x(-1)}{3}, \overset{x(0)}{-3}, \overset{x(1)}{0}, \overset{x(2)}{4}, \overset{x(3)}{-2} \}$$

$$- X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-2}^3 x(n) z^{-n}$$

$$= x(-2)z^2 + x(-1)z^1 + x(0)z^0 + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3}$$

$$= 5z^2 + 3z^1 - 3z^0 + 0z^{-1} + 4z^{-2} + (-2)z^{-3}$$

$$X(z) = 5z^2 + 3z^1 - 3 + 4z^{-2} - 2z^{-3}$$

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A Finite duration sequence $x(n) = \{5, 3, 0, 1, 2, 4\}$
 Find Z-transform of $x(n)$.

$$- x(n) = \begin{matrix} & x(0) & x(1) & x(2) & x(3) & x(4) & x(5) \\ \{ & 5 & 3 & 0 & 1 & 2 & 4 \end{matrix}$$

$$- X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=0}^5 x(n) z^{-n}$$

$$= x(0)z^0 + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + x(4)z^{-4} + x(5)z^{-5}$$

$$= 5z^0 + 3z^{-1} + 0z^{-2} + 1z^{-3} + 2z^{-4} + 4z^{-5}$$

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2. Z Transform Examples

A Finite duration Sequence $x(n) = \{5, 3, 0, 1, 2, 4\}$
Find Z-transform of $x(n)$.

$$x(n) = \{ \overset{x(0)}{5}, \overset{x(1)}{3}, \overset{x(2)}{0}, \overset{x(3)}{1}, \overset{x(4)}{2}, \overset{x(5)}{4} \}$$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=0}^5 x(n) z^{-n} \\ &= x(0)z^0 + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + x(4)z^{-4} + x(5)z^{-5} \\ &= 5z^0 + 3z^{-1} + 0z^{-2} + 1z^{-3} + 2z^{-4} + 4z^{-5} \\ &= \boxed{5 + 3z^{-1} + z^{-3} + 2z^{-4} + 4z^{-5}} \end{aligned}$$

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2. Z Transform Examples:

Examples of Inverse Z transform

Find the Signal $x(n)$ for which the Z transform is

$$X(z) = 4z^4 - z^3 - 3z + 4z^{-1} + 3z^{-2}$$

$$- \quad x(n) = Z^{-1}[X(z)] = \frac{1}{2\pi j} \int X(z) z^{n-1} dz$$

$$X(z) = \underset{\substack{\uparrow \\ x(-4)}}{4} z^4 - \underset{\substack{\uparrow \\ x(-3)}}{1} z^3 + \underset{\substack{\uparrow \\ x(-2)}}{0} z^2 - \underset{\substack{\uparrow \\ x(-1)}}{3} z + \underset{\substack{\downarrow \\ \uparrow \\ x(0)}}{0} z^0 + \underset{\substack{\uparrow \\ x(1)}}{4} z^{-1} + \underset{\substack{\uparrow \\ x(2)}}{3} z^{-2}$$

$$- \quad X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \\ = \sum_{n=-4}^2 x(n) z^{-n}$$

$$- \quad x(n) = \{ 4, -1, 0, -3, 0, 4, 3 \}$$

$$= \sum_{n=-4}^{\infty} x(n) z^{-n}$$

$$- \quad x(n) = \{ 4, -1, 0, -3, 0, 4, 3 \}$$

If $x(z) = 2z^2 + 3$, find $x(n)$

$$- \quad x(z) = 2z^2 + 0 \times z^1 + 3 \times z^0$$

$$- \quad x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-2}^0 x(n) z^{-n}$$

$$- \quad x(n) = \{ 2, 0, 3 \}$$

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Example of Z Transform for standard basic signal

7:52

1. Z Transform Examples: Standard Basic Signal in Sig...
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$$\begin{aligned}
 &= \sum_{n=-\infty}^{\infty} x(n) z^{-n} \\
 &- x(n) = \{2, 0, 3\} \\
 &\text{If } x(z) = 3 + 2z^{-1} + 3z^{-3}, \text{ find } x(n) \\
 &- x(z) = \frac{3 \times z^0}{x(0)} + \frac{2 \times z^{-1}}{x(1)} + \frac{0 \times z^{-2}}{x(2)} + \frac{3 \times z^{-3}}{x(3)} \\
 &- x(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=0}^3 x(n) z^{-n} \\
 &- x(n) = \{3, 2, 0, 3\}
 \end{aligned}$$

Z Transform in Signal and ... · 4/44

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Inverse Z Transform

15:06

Z Transform and Inverse Z Transform: Relation with Di...
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Z Transform for finite discrete sequence

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Examples of Inverse Z Transform

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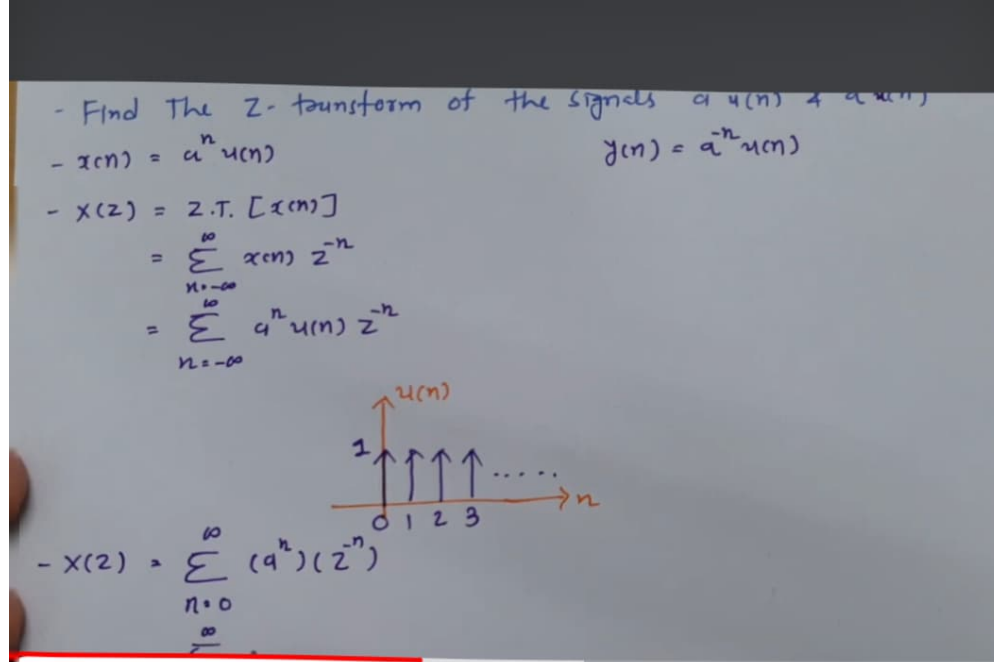
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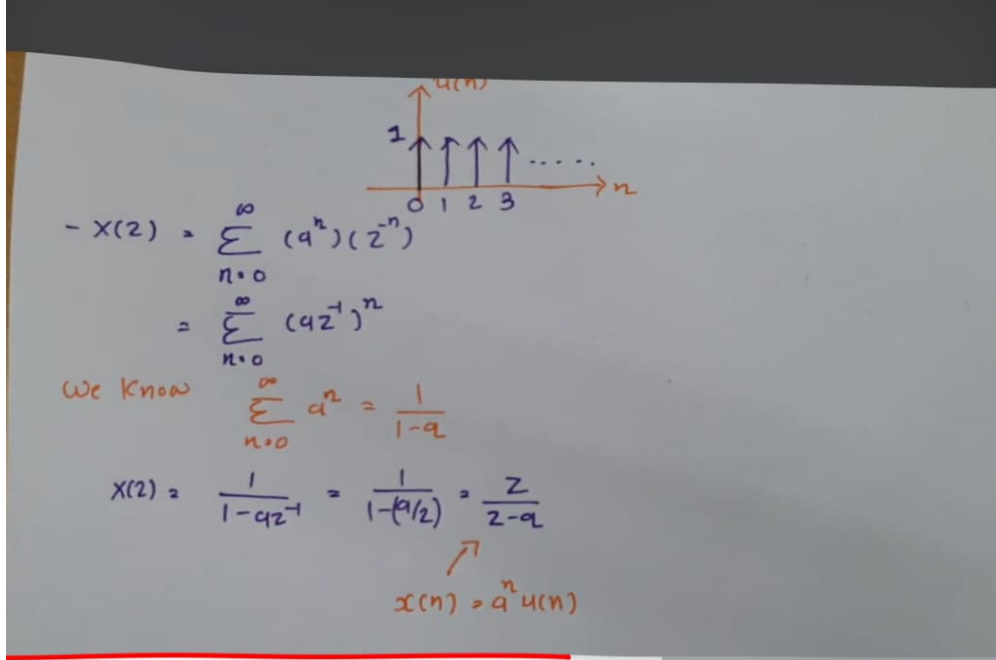
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Z Transform of Unit Step



$$\begin{aligned}
 X(z) &= \sum_{n=0}^{\infty} (a^n)(z^{-n}) \\
 &= \sum_{n=0}^{\infty} (az^{-1})^n
 \end{aligned}$$

We know $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$

$$X(z) = \frac{1}{1-az^{-1}} = \frac{1}{1-a/2} = \frac{z}{z-a}$$

$x(n) = a^n u(n)$

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Z Transform of Unit Step Signal

3:01

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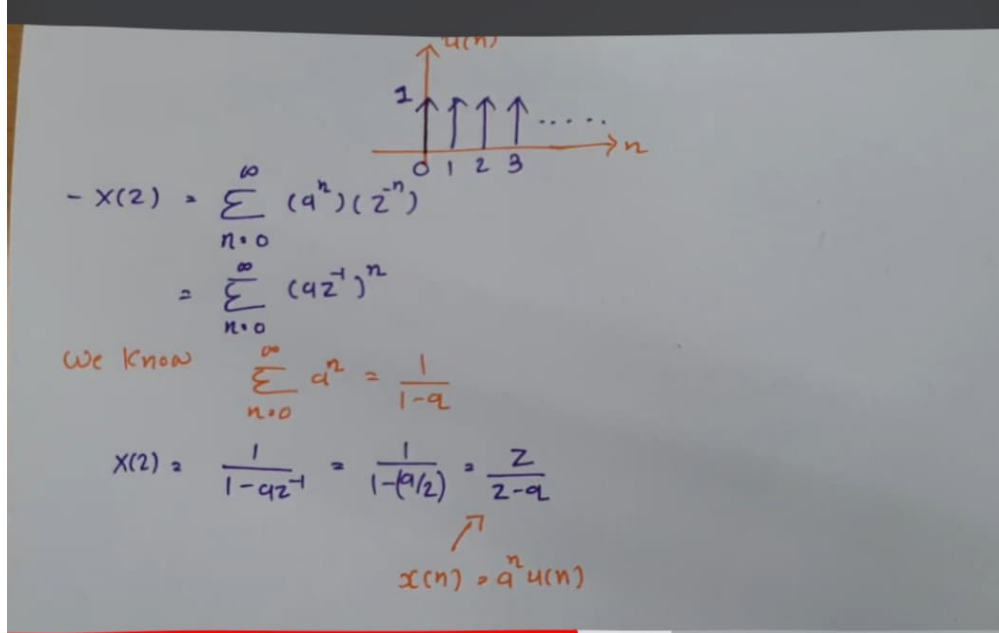
Z Transform of Unit Step Signal

3:13

Z Transform of Unit Impulse Signal in Signals and Syste...

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$$X(z) = \sum_{n=0}^{\infty} (a^n)(z^{-n})$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

We know $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$

$$X(z) = \frac{1}{1-az^{-1}} = \frac{1}{1-a/2} = \frac{z}{z-a}$$

$x(n) = a^n u(n)$

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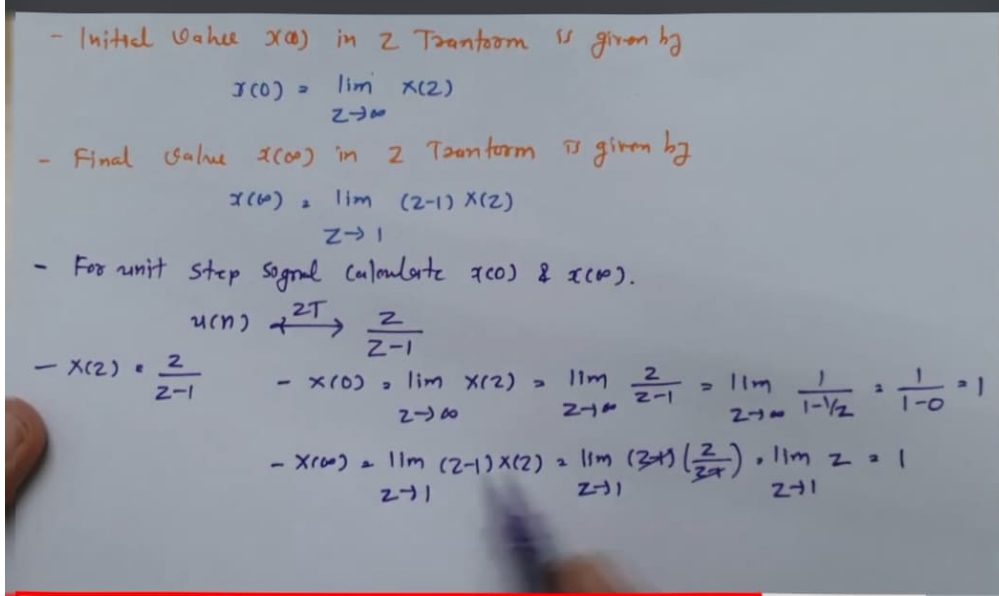
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Z Transform of Cosine Signal 10:31

2. Z Transform of Cosine Signal in Signals and System

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Z Transform of Fundamental 10:05

Z Transform of Fundamental Signals in Signals and Syste...

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Initial value theorem & Final value theorem in Z Transform

Initial and Final Value Theor...

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Example of Initial value theorem in Z Transform 3:23

Initial Value Theorem in Z Transform with Solved Ex...

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Example of Final value theorem in Z Transform 6:15

Final Value Theorem in Z Transform with Solved Exa...

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Properties of Z Transform

- Initial Value $x(0)$ in Z Transform is given by

$$x(0) = \lim_{z \rightarrow \infty} x(z)$$

- Final Value $x(\infty)$ in Z Transform is given by

$$x(\infty) = \lim_{z \rightarrow 1} (z-1)x(z)$$

- For unit step signal calculate $x(0)$ & $x(\infty)$.

$u(n) \xrightarrow{ZT} \frac{z}{z-1}$

- $x(z) = \frac{z}{z-1}$

- $x(0) = \lim_{z \rightarrow \infty} x(z) = \lim_{z \rightarrow \infty} \frac{z}{z-1} = \lim_{z \rightarrow \infty} \frac{1}{1-1/z} = \frac{1}{1-0} = 1$

- $x(\infty) = \lim_{z \rightarrow 1} (z-1)x(z) = \lim_{z \rightarrow 1} (z-1)\left(\frac{z}{z-1}\right) = \lim_{z \rightarrow 1} z = 1$

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Z Transform of Fundamentals 10:05

Signals in Signals and Syste...

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Initial value theorem & Final value theorem in Z Transform 11:00

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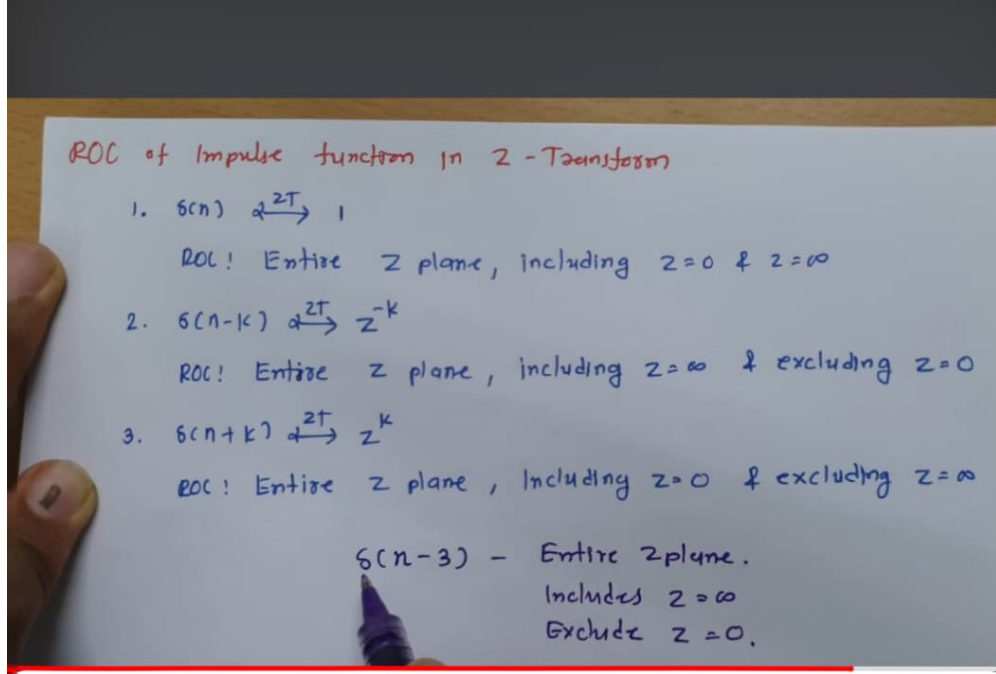
Properties of Z Transform 6:11

Properties of Z Transform Explained: Linearity, Time S...

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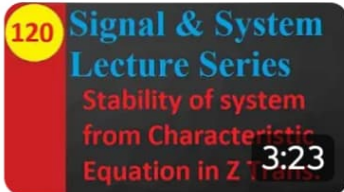


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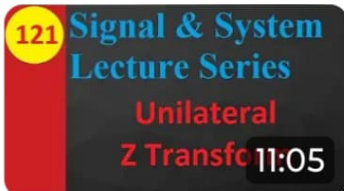


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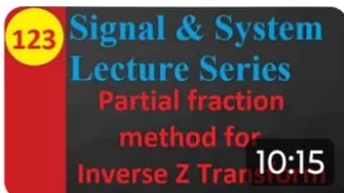
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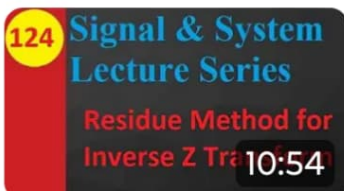
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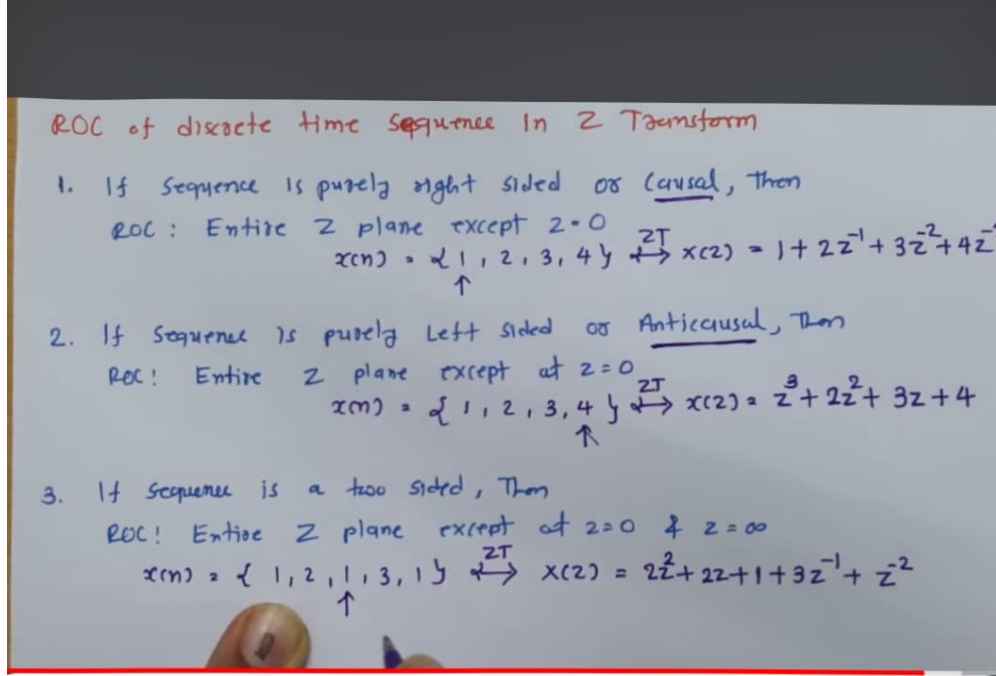
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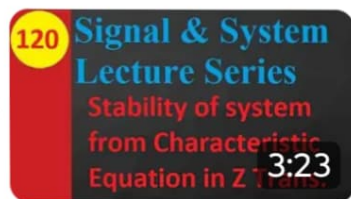
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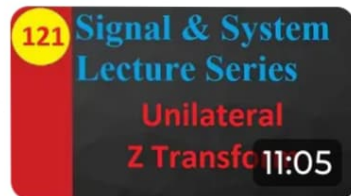
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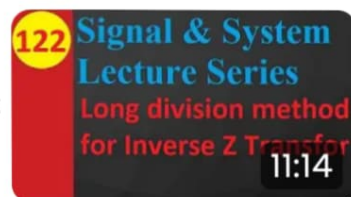
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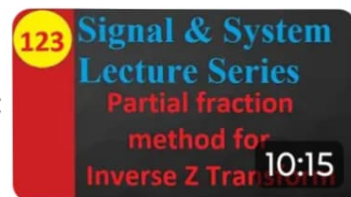
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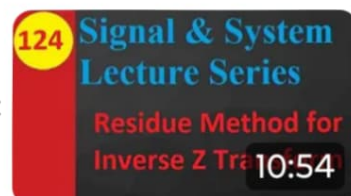
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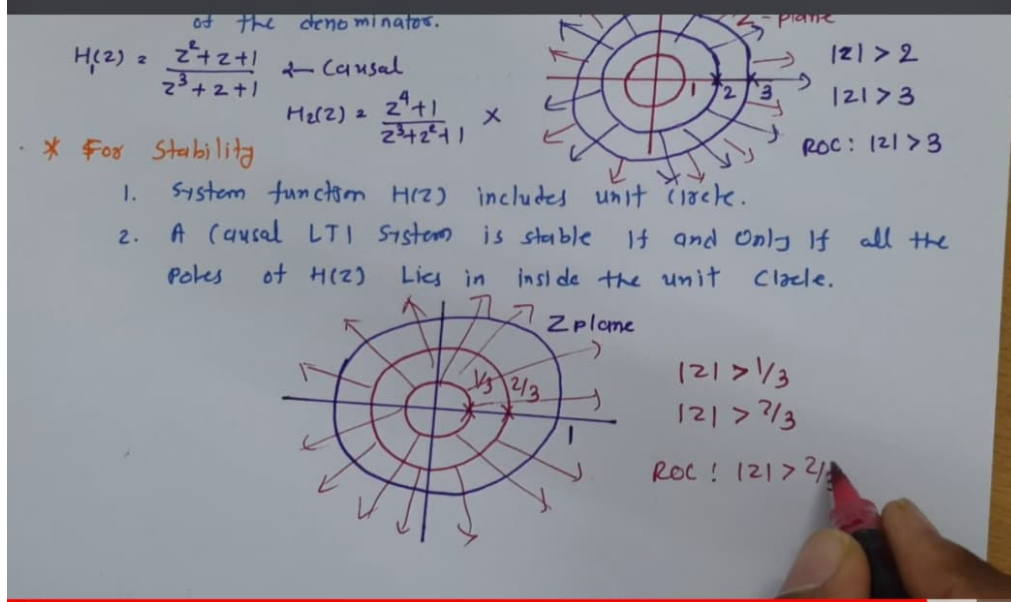
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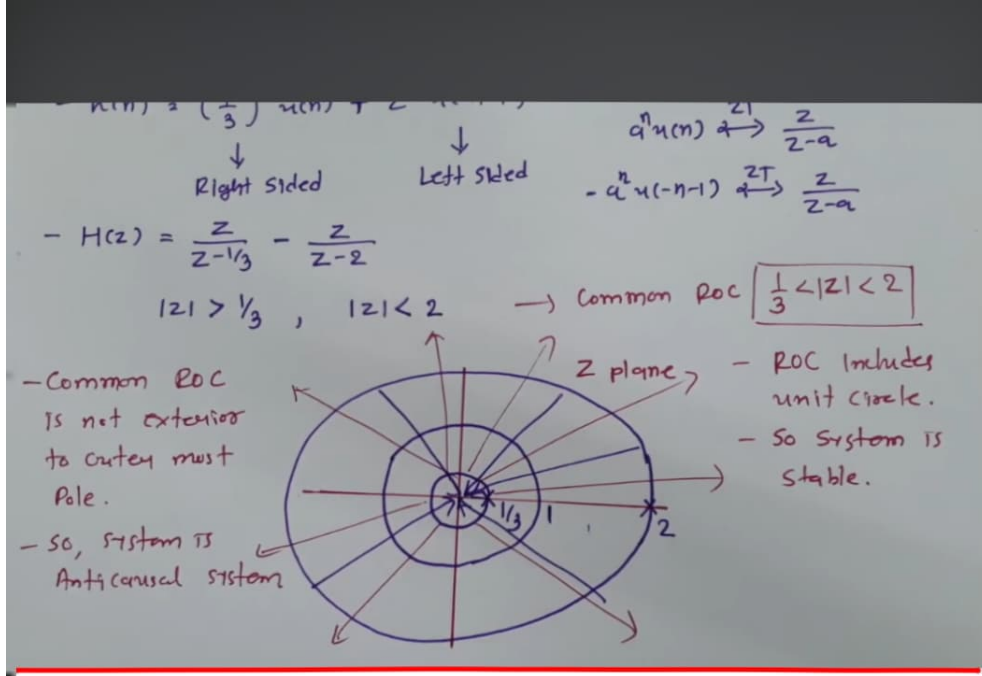
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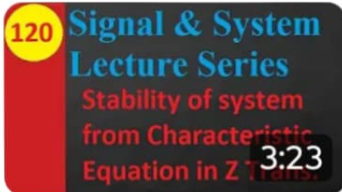


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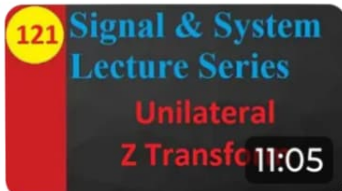


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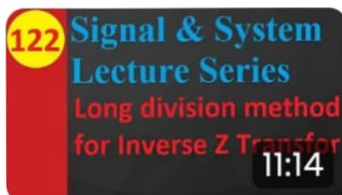
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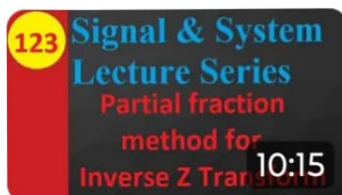
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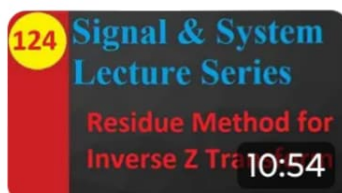
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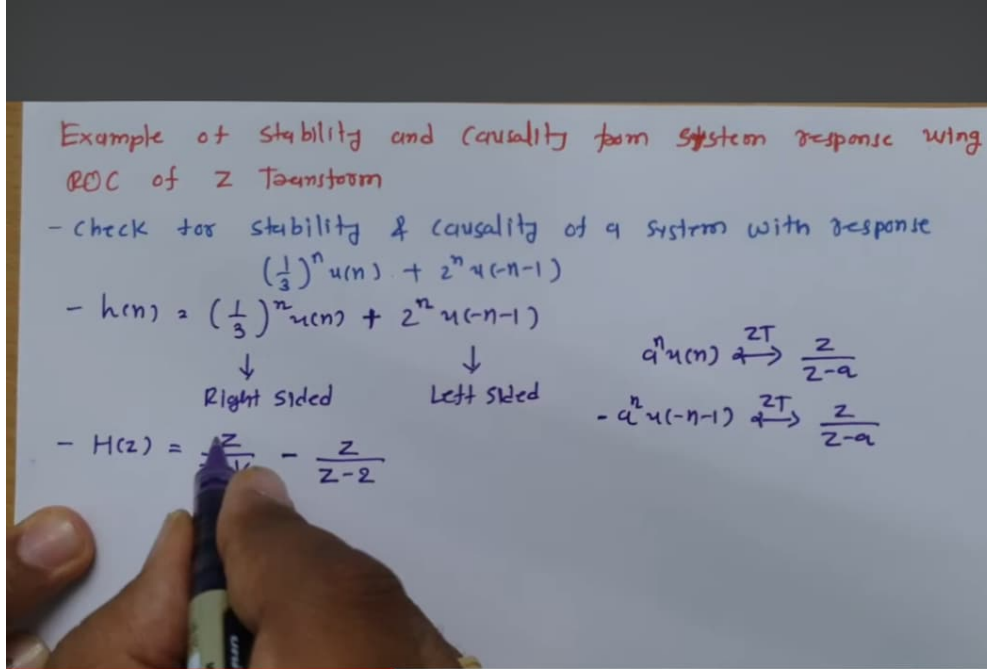
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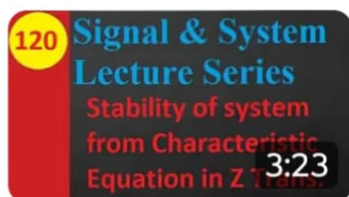
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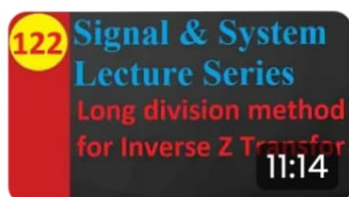
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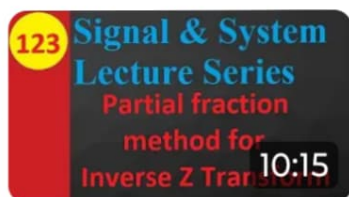
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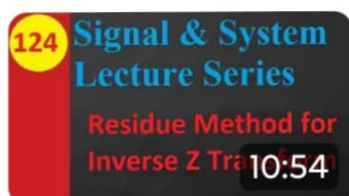
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- e.g. Find VZT $x(n) = \{3, 4, 5, 0, 1, 2\}$

$\times \begin{matrix} -2 & -1 & 0 & 1 & 2 & 3 \\ \hline & & & & \checkmark & \end{matrix}$

$$\text{VZT } [x(n)] = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$= x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + \dots$$

$$= 5 + 0 \times z^{-1} + 1 \times z^{-2} + 2 \times z^{-3}$$

$$= 5 + z^{-2} + 2z^{-3}$$

$$\text{ZT } [x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= x(-2)z^2 + x(-1)z^1 + x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3}$$

$$= 3z^2 + 4z + 5 + z^{-2} + 2z^{-3}$$

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Q

$$x(n) = u(n)$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} u(n) e^{-j\omega n}$$

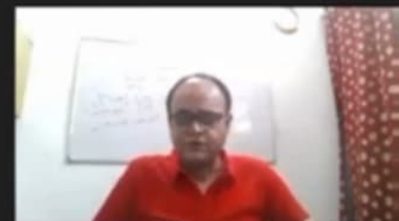
$$= \sum_{n=0}^{\infty} 1 \times (e^{-j\omega})^n$$

$$= 1 + e^{-j\omega} + e^{-2j\omega} + e^{-3j\omega} + \dots \infty$$

$$X(\omega) = \frac{1}{1 - e^{-j\omega}} \rightarrow \omega = 0$$

$$X(\omega) = \infty \quad (X)$$

$$u(n) = 1 \quad \forall n \geq 0$$

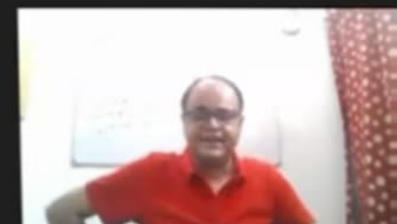


$$X(\omega) = \frac{1}{e^{j\omega/2} \cdot e^{-j\omega/2} - e^{-j\omega/2} \cdot e^{j\omega/2}}$$

$$= \frac{1}{e^{-j\omega/2} \left[e^{j\omega/2} - e^{-j\omega/2} \right]}$$

$$\frac{e^{j\omega/2} - e^{-j\omega/2}}{2j} = \sin\left(\frac{\omega}{2}\right)$$

$$= \frac{e^{j\omega/2}}{2j \sin \omega/2}$$



$$X(\omega) = \frac{1}{e^{j\omega/2} \cdot e^{-j\omega/2} - e^{-j\omega/2} \cdot e^{j\omega/2}} = \frac{1}{e^{-j\omega/2} \left[e^{j\omega/2} - e^{-j\omega/2} \right]}$$

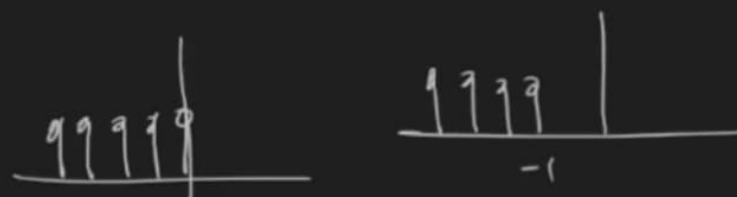
$$\frac{e^{j\omega/2} - e^{-j\omega/2}}{2j} = \sin\left(\frac{\omega}{2}\right)$$

$$= \frac{e^{j\omega/2}}{2j \sin \omega/2}$$

$$\boxed{u(n) \xleftrightarrow{\text{DTFT}} \frac{e^{j\omega/2}}{2j \sin \omega/2}}$$



Q: $x(n) = a^{-n} u(-n-1)$



$$\sum_{n=-\infty}^{-1} a^{-n} e^{j\omega n}$$

✓ $n = -m$

$$= \sum_{m=1}^{\infty} (a e^{j\omega})^m = \checkmark$$



$$X(\omega) = A \left[\frac{(e^{-j\omega})^0 - (e^{-j\omega})^{L-1+1}}{1 - e^{-j\omega}} \right]$$

$$X(\omega) = A \left[\frac{1 - e^{-j\omega L}}{1 - e^{-j\omega}} \right] = A \left[\frac{e^{j\omega L/2} e^{-j\omega L/2} - e^{-j\omega L/2} e^{j\omega L/2}}{e^{-j\omega/2} e^{j\omega/2} - e^{j\omega/2} e^{-j\omega/2}} \right]$$

$$X(\omega) = A \frac{e^{-j\omega L/2}}{e^{-j\omega/2}} \left[\frac{e^{j\omega L/2} - e^{-j\omega L/2}}{e^{j\omega/2} - e^{-j\omega/2}} \right]$$

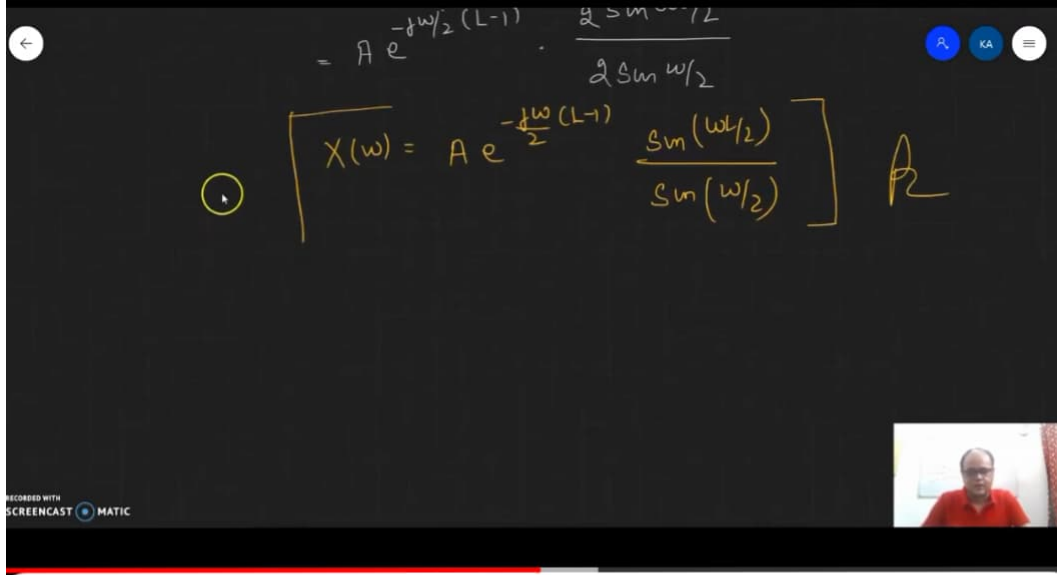


$$X(\omega) = A \left[\frac{1}{1 - e^{-j\omega}} \right] \left[e^{-j\omega/2} e^{j\omega/2} - e^{-j\omega/2} e^{-j\omega/2} \right]$$

$$X(\omega) = A \left(\frac{e^{-j\omega L/2}}{e^{-j\omega/2}} \right) \left[\frac{e^{j\omega/2} - e^{-j\omega L/2}}{e^{j\omega/2} - e^{-j\omega/2}} \right]$$

$$= A e^{-j\omega/2(L-1)} \cdot \frac{2 \sin \omega L/2}{2 \sin \omega/2}$$





Discrete Time Fourier Transform [Properties & Frequency Domain Analysis..

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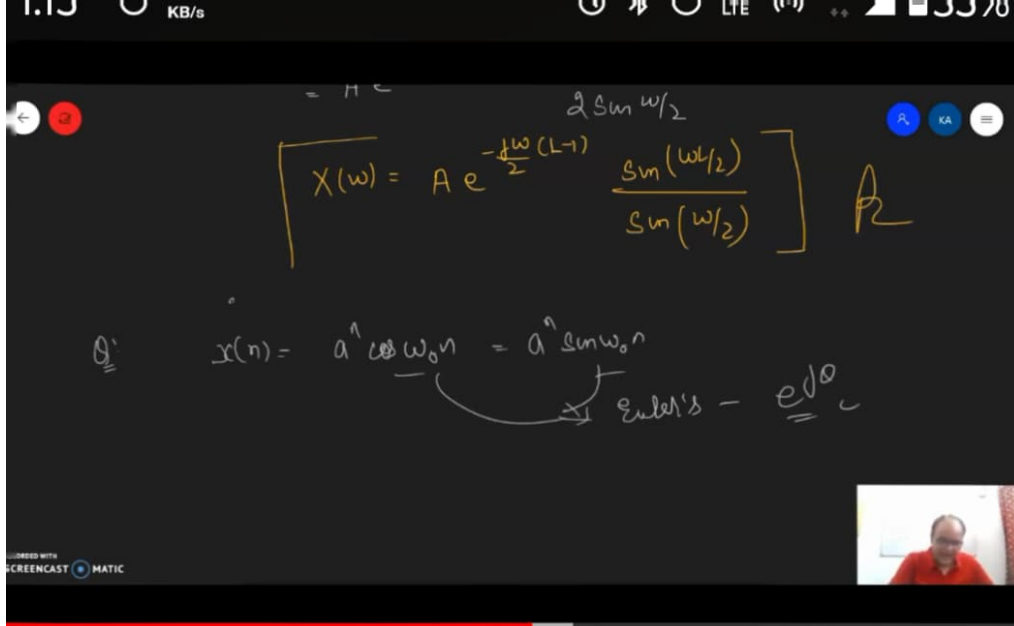
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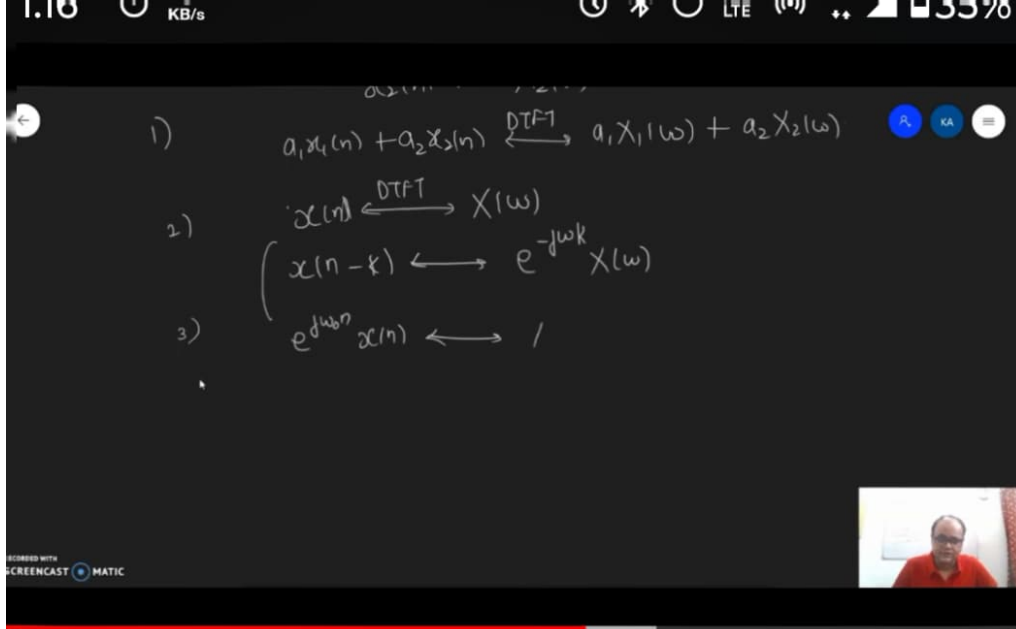
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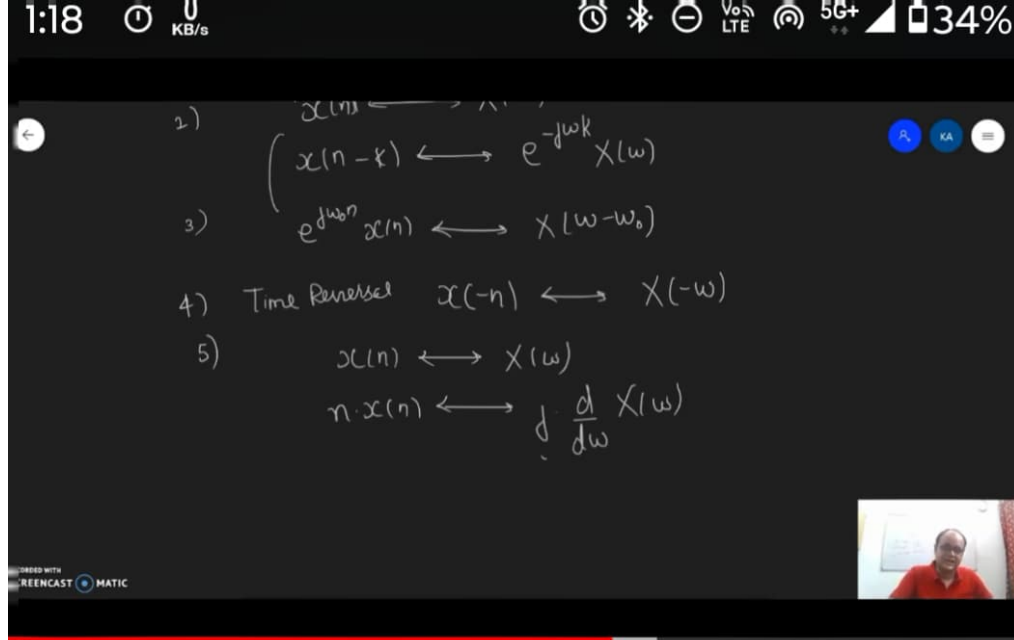
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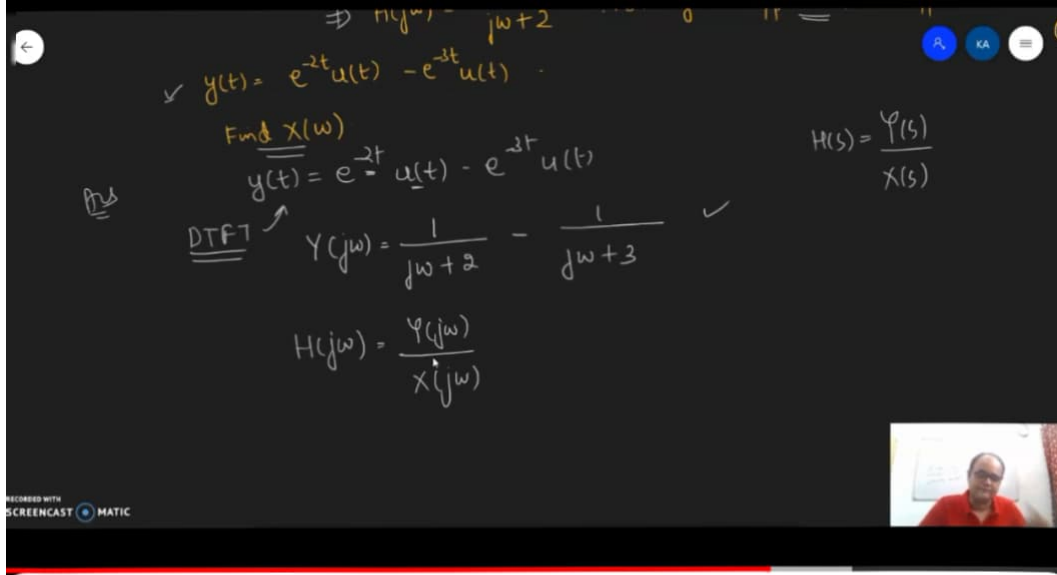
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Energy & Power

23:40

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Handwritten mathematical derivations for Digital Signal Processing:

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} \Rightarrow X(j\omega) = \frac{Y(j\omega)}{H(j\omega)}$$

$$X(j\omega) = \left(\frac{1}{j\omega+2} - \frac{1}{j\omega+3} \right) \cdot (j\omega+2)$$

$$X(j\omega) = 1 - \frac{j\omega+2}{j\omega+3} = \text{IDFT} \left[1 \right] + \text{IDFT} \left[- \right]$$

$$x(n) = \text{IDFT} \left[X(j\omega) \right]$$

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Signals & Systems

Unit-I



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Course · 103 lessons

Handwritten mathematical derivation on a blackboard background:

$$X(\omega) = \frac{1}{j\omega+2} - \frac{1}{j\omega+3}$$

$$x(n) = \text{IDFT}[X(\omega)] = \text{IDFT}\left[\frac{1}{j\omega+2}\right] - \text{IDFT}\left[\frac{1}{j\omega+3}\right]$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{j\omega n}}{j\omega+2} d\omega - \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{j\omega n}}{j\omega+3} d\omega$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{\pi} \frac{e^{j\omega n}}{j\omega+2} d\omega - \int_{-\pi}^{\pi} \frac{e^{j\omega n}}{j\omega+3} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\frac{e^{+n} - e^{-n}}{j} \right]$$

Small video inset of a person in a red shirt is visible in the bottom right corner of the video frame.

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$$X(\omega) = -\frac{j\omega + 2}{j\omega + 3}$$

$$\frac{X(\omega)}{j\omega} = \frac{-j\omega + 2}{(j\omega)(j\omega + 3)} = \frac{A_1}{j\omega} + \frac{A_2}{j\omega + 3}$$

Small video inset showing a person in a red shirt.

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$$X(\omega) = \checkmark X(j\omega) = 1 - \frac{j\omega + 2}{j\omega + 3} = \cancel{\text{IDFT} \left[1 \right]} + \text{IDFT} \left[\frac{-j\omega + 1}{j\omega + 3} \right]$$

$\swarrow$
0

$x(n)$

$$x(n) = \text{IDFT} [X(\omega)]$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 e^{j\omega n} d\omega = \frac{1}{2\pi}$$



$$\frac{X(w)}{e^{jw}} = \frac{A_1}{e^{jw} - \frac{1}{2}} + \frac{A_2}{e^{jw} - \frac{1}{3}}$$

$$X(w) = \frac{3e^{jw}}{e^{jw} - \frac{1}{2}} - \frac{2e^{jw}}{e^{jw} - \frac{1}{3}}$$

$$A_1 = 3$$
$$A_2 = -2$$

A

KA

≡



$$X(w) = \frac{1}{\left(e^{jw} - \frac{1}{2}\right)\left(e^{jw} - \frac{1}{3}\right)}$$

$$\frac{X(w)}{e^{jw}} = \frac{A_1}{e^{jw} - \frac{1}{2}} + \frac{A_2}{e^{jw} - \frac{1}{3}}$$

$$\begin{aligned} A_1 &= 3 \\ A_2 &= -2 \end{aligned}$$

$$X(w) = \frac{3e^{jw}}{e^{jw} - \frac{1}{2}} - \frac{2e^{jw}}{e^{jw} - \frac{1}{3}}$$

$$X(w) = \frac{3}{1 - \frac{1}{2}e^{-jw}} - \frac{2}{1 - \frac{1}{3}e^{-jw}}$$

$$= 3\left(\frac{1}{2}\right)^n u(n) - 2\left(\frac{1}{3}\right)^n u(n)$$



$$\frac{X(\omega)}{e^{j\omega}} = \frac{A_1}{e^{j\omega} - \frac{1}{2}} + \frac{A_2}{e^{j\omega} - \frac{1}{3}}$$

$$\begin{aligned} A_1 &= 3 \\ A_2 &= -2 \end{aligned}$$

$$X(\omega) = \frac{3e^{j\omega}}{e^{j\omega} - \frac{1}{2}} - \frac{2e^{j\omega}}{e^{j\omega} - \frac{1}{3}}$$

$$X(\omega) = \frac{3}{1 - \frac{1}{2}e^{-j\omega}} - \frac{2}{1 - \frac{1}{3}e^{-j\omega}}$$

$$= 3\left(\frac{1}{2}\right)^n u(n) - 2\left(\frac{1}{3}\right)^n u(n)$$



$$\checkmark \quad E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t) \cdot \overbrace{x^*(t)}^{dt}$$

$$x(t) = \text{IFT } X(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$x^*(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega$$

$$E = \int_{-\infty}^{\infty} x(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega dt$$

$$x^* = \alpha - j\beta$$

$$x \cdot x^* = (\alpha + j\beta)(\alpha - j\beta) \\ = \alpha^2 + \beta^2$$

$$x x^* = |x|^2$$

$x \rightarrow$ Real valued signal

$$x = x^*$$

$$t \Rightarrow 0$$



$$\checkmark \quad E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t) \cdot \overbrace{x^*(t)} dt$$

$$x(t) = \text{IFT } X(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$x^*(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega$$

$$E = \int_{-\infty}^{\infty} x(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega dt$$

$$x^* = \alpha - j\beta$$

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$$t \rightarrow 0$$



$$E = \int_{-\infty}^{\infty} x(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega dt$$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} X^*(\omega) d\omega dt$$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{X^*(\omega)}_{\sim} \underbrace{X(\omega)}_{\sim} d\omega$$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) \left[\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right] d\omega$$

$\downarrow$
 $X(\omega)$



$$E = \int_{-\infty}^{\infty} x(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega dt$$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} X^*(\omega) d\omega dt$$

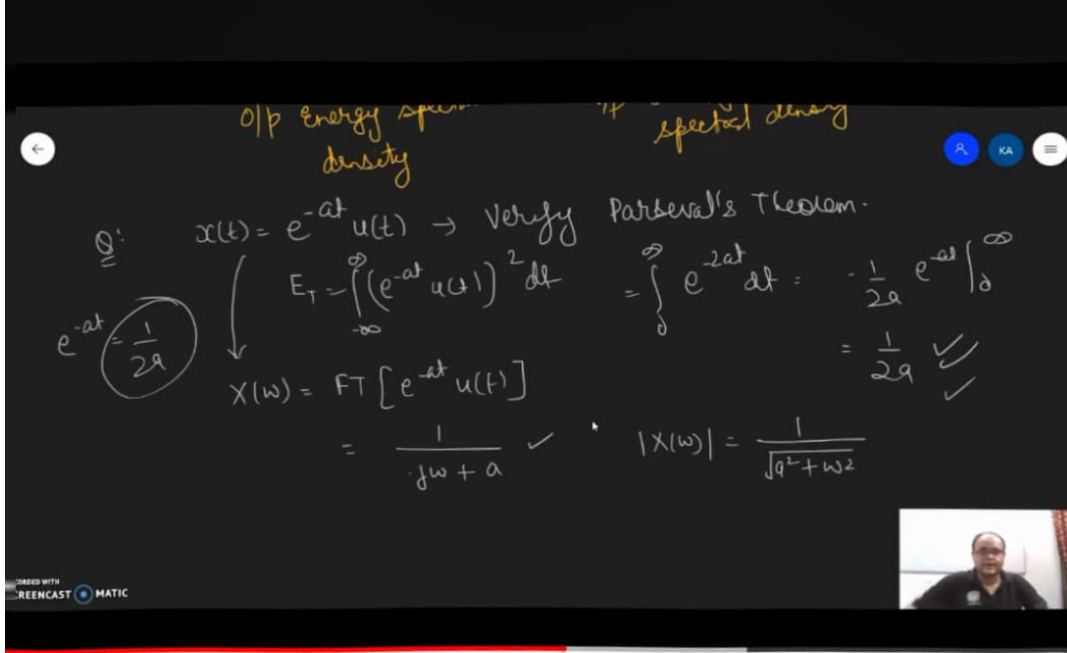
$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) \left[\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right] d\omega$$

$\downarrow$
 $X(\omega)$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{X^*(\omega)} \underbrace{X(\omega)} d\omega$$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |x(t)|^2 dt$$





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Parseval's Theorems [Energy and Power Calculation in Frequency Domain]

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Handwritten derivation on a blackboard background:

$$e^{-at} = \frac{1}{2a}$$

$$X(\omega) = \text{FT} [e^{-at} u(t)] = \frac{1}{j\omega + a}$$

$$|X(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}$$

$$|X(\omega)|^2 = \frac{1}{\omega^2 + a^2}$$

$$E_F = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + a^2} d\omega = \frac{1}{2\pi a} \left(\tan^{-1} \left(\frac{\omega}{a} \right) \right) \Big|_{-\infty}^{\infty}$$

$$= \frac{1}{2\pi a} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = \frac{1}{2a}$$

Arrows indicate the relationship between the time domain signal, its Fourier transform, the magnitude squared, and the final energy calculation.

Description

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$$P = \sum_{-\infty}^{\infty} |C_n|^2$$

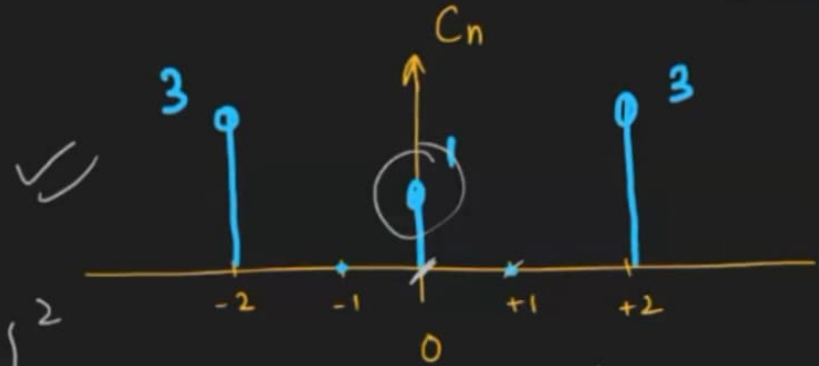
$$= |C_{-2}|^2 + |C_{-1}|^2 + |C_0|^2 + |C_1|^2 + |C_2|^2$$

$$= 3^2 + 0^2 + 1^2 + 0^2 + 3^2$$

$$= 9 + 1 + 9$$

$$= 19$$

Q:



$$= 1 + 3 \left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right)$$

$$= 1 + 6 \cos 2\omega_0 t$$

$$= 1 + 18 = 19$$

$$A^2/2$$

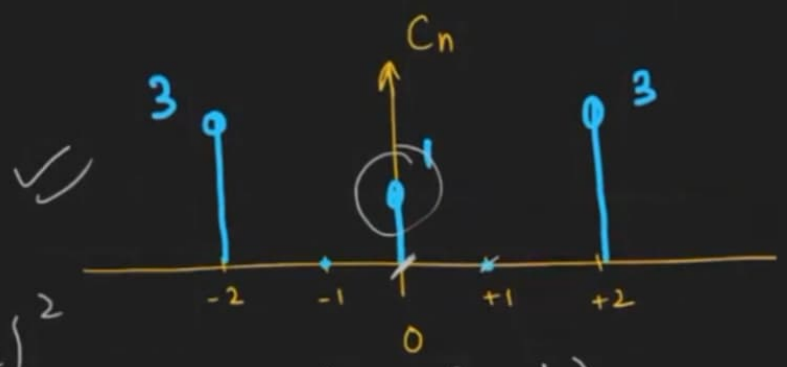
$$= C_{-2} e^{-j\omega_0 t}$$

$$= 3e$$



n

Q:



Power = ?

$$x[n] = \sum C_n e^{jn\omega_0 t}$$

$$+ |C_{-1}|^2 + |C_0|^2 + |C_1|^2 + |C_2|^2$$

$$0^2 + 1^2 + 0^2 + 3^2$$

$$= 1 + 3 \left(e^{2j\omega_0 t} + e^{-2j\omega_0 t} \right)$$

$$= C_{-2} e^{-2j\omega_0 t} + 0 + C_0 + 0 + C_2 e^{2j\omega_0 t}$$

$$= 3 e^{-2j\omega_0 t} + 1 + 3 e^{2j\omega_0 t}$$

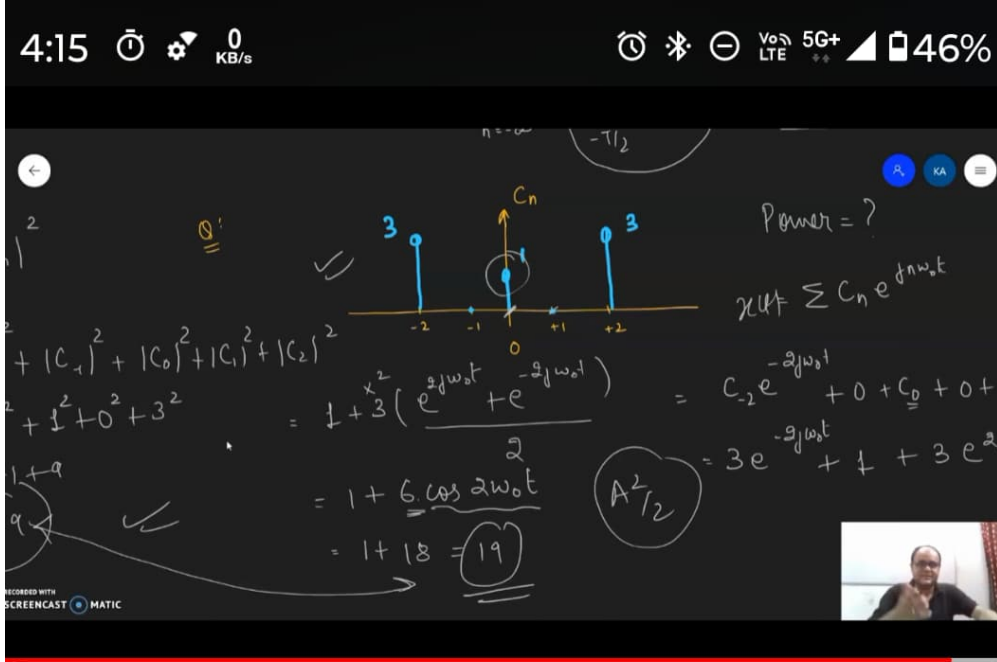
$$= 1 + 6 \cos 2\omega_0 t$$

$A^2/2$

$$= 1 + 18 = 19$$

19





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SIGNALS & SYSTEMS

Cross-Correlation and Auto-Correlation

Properties Explained

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Next: Eigen Functions and Eigen Values

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Sampling Theorem

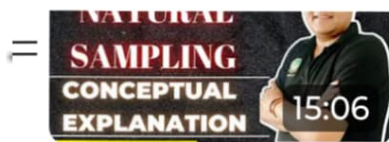
A bandlimited signal can be represented in its samples and can be easily reconstructed if the sampling frequency is greater than or equal to twice the maximum frequency component present in the signal.

$$\omega_s \geq 2\omega_m \quad \text{or} \quad f_s \geq 2f_m$$

$x(t)$ Bandlimited $\times$ $X(\omega)$ FT of $x(t)$

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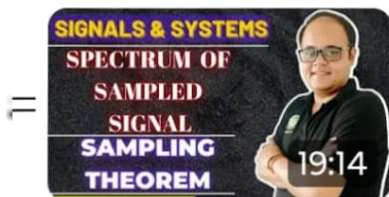
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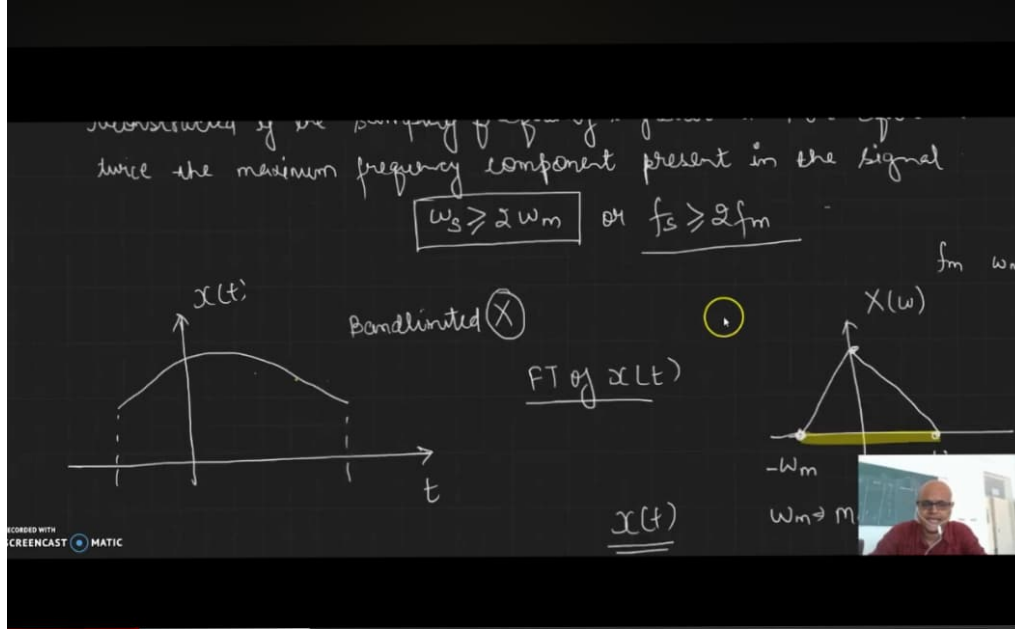


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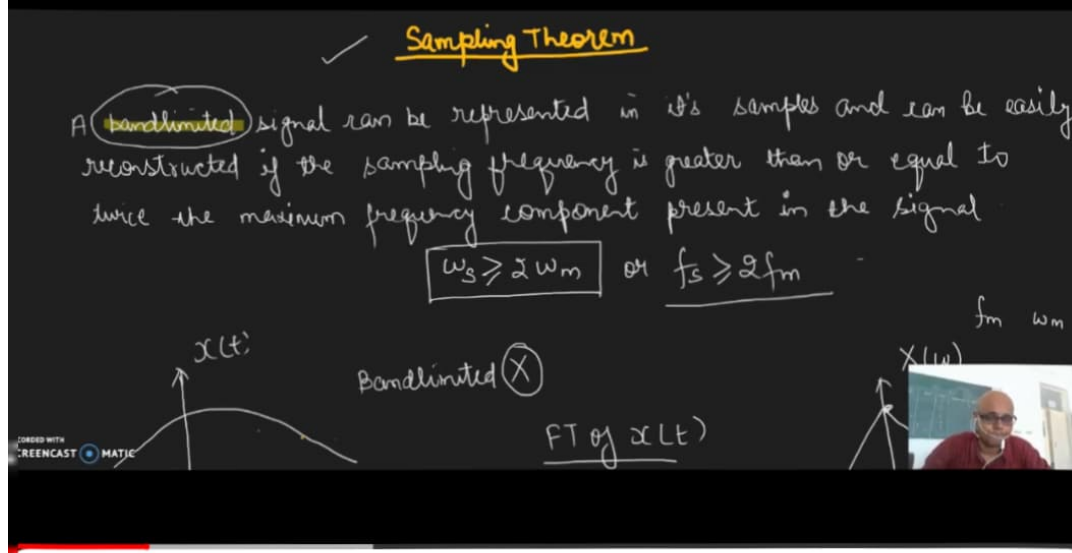


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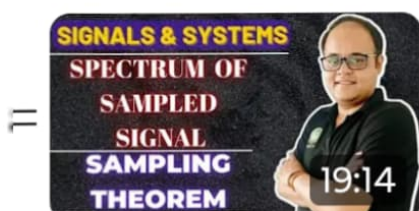
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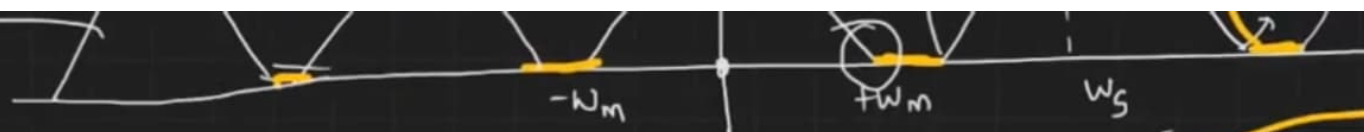
$$\begin{aligned}
 S(\omega) &= \frac{1}{2\pi} \left[X(\omega) * \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s) \right] \\
 &= \frac{\omega_s}{2\pi} \left[X(\omega) * \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s) \right] \\
 &= \frac{1}{T_s} \left[\sum_{n=-\infty}^{\infty} X(\omega) * \delta(\omega - n\omega_s) \right] \checkmark \\
 &= \frac{1}{T_s} \left[\sum_{n=-\infty}^{\infty} X(\omega - n\omega_s) \right] \checkmark
 \end{aligned}$$

$\omega_s = \frac{2\pi}{T_s}$

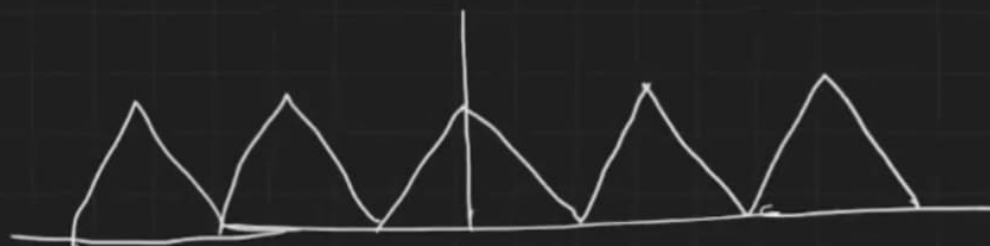
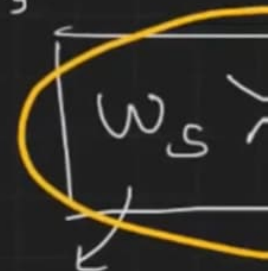
सिंपल हम कर चुके कंडीशन में बदल सकते हैं



oversampling

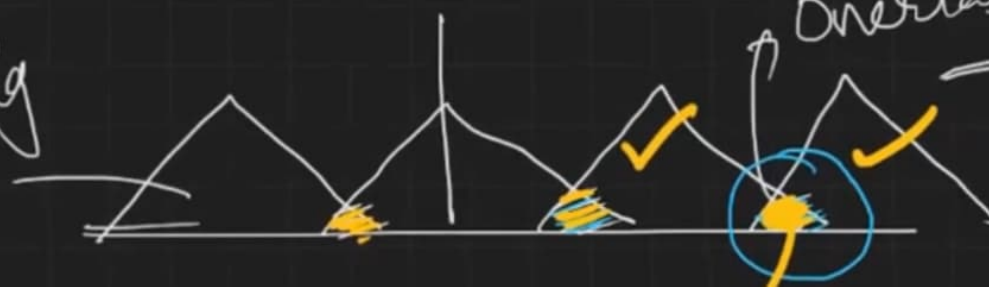


$$W_s - W_m > W_m$$



$$\leftarrow W_s = 2W_m$$

undersampling



overlapping



Nyquist
Criteria

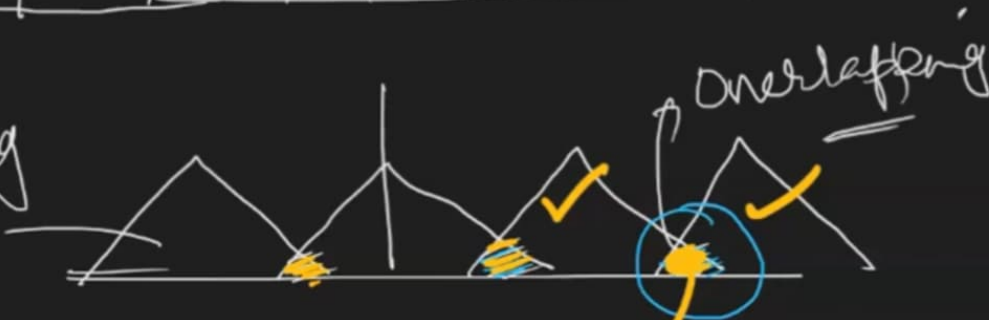
oversampling

$\Rightarrow f_s = 2f_m$ $\omega_s = 2\omega_m$ $\omega_s - \omega_m > \omega_m$

$\omega_s >$

$\omega_s = 2\omega_m$

undersampling



Properties:-

- 1) $\checkmark$ $x(t) \xrightarrow{NR} f_s$ then $x(t \pm t_0) \xrightarrow{NR} f_s$
- 2) $x(t) \xrightarrow{NR} f_s$ then $x(at) \xrightarrow{NR} a \cdot f_s$
- 3) $x(t) \xrightarrow{NR} f_s$ then $[x(t)]^n \xrightarrow{NR} n \cdot f_s$
- 4) $x(t) \xrightarrow{NR} f_s$ then $\frac{d}{dt} x(t) \xrightarrow{NR} f_s$
- 5) $x(t) \xrightarrow{NR} f_s$ then $\int x(t) dt \xrightarrow{NR} f_s$



4) $x(t) \longrightarrow f_s$ then $\frac{d}{dt} x(t) \longrightarrow f_s$

5) $x(t) \xrightarrow{NR} f_s$ then $\int x(t) dt \xrightarrow{NR} f_s$

6) $x_1(t) \xrightarrow{NR} f_{s_1}$ then $x(t) = x_1(t) \cdot x_2(t) \xrightarrow{NR} f_s = f_{s_1} + f_{s_2}$
 $x_2(t) \xrightarrow{NR} f_{s_2}$



$$x_2(t) \longrightarrow \omega_{s_2}$$

multiply

Q: $x(t) = 10 \sin 6\pi t + \cos 2\pi t$

$$x_1(t)$$

$$x_2(t)$$

$$\omega_{s_1} = 6\pi$$

$$\omega_{s_2} = 2\pi$$

$$\omega_m = \omega_{s_1}$$

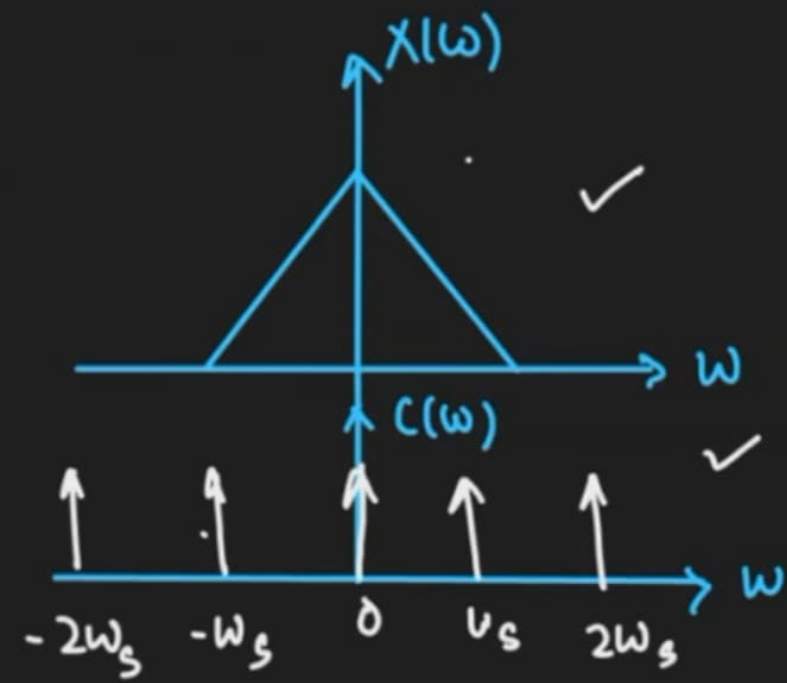
$$\omega_m = 6\pi$$

$$\omega_s = 2\omega_m = 12\pi$$

$$f_s = \frac{\omega_s}{2\pi} = 6 \text{ Hz}$$

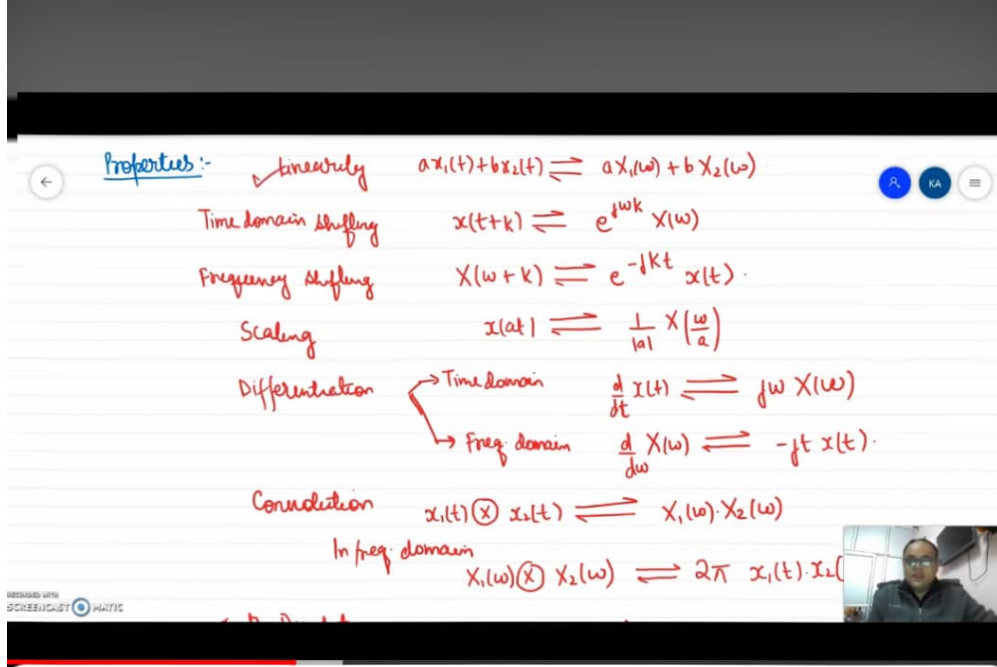
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$$X(\omega) * \delta(\omega - \omega_s) = X(\omega - \omega_s)$$



Fourier Transform-Basic Introduction [Definition, Existence, Fourier Transfor...

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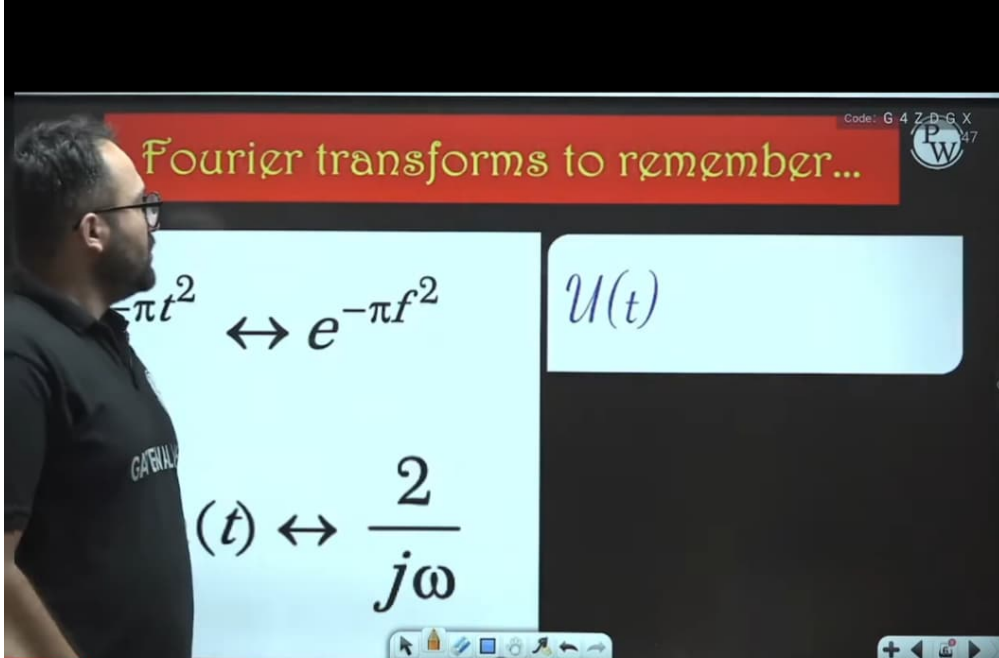
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Two method to find th CTFT

Energy Signal.

Find L.T of Signal and
replace $s \rightarrow j\omega$

Power Signals.

→ we will not apply
Formulae
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Duality theorem...

$a < 0$
 $e^{at} u(t) \rightarrow \frac{1}{s-a}, \sigma > a. \text{ ROC include:}$

$e^{-(at)} u(t)$ and duality

$(e^{at} u(t) \rightarrow \frac{1}{j\omega - a})$
 $\frac{1}{j\omega - a} \rightarrow 2\pi e^{-a\omega} u(-\omega)$

THE WALLAH

...

Fourier transforms to remove...

Code: G 4 Z-D-G X



$$e^{-\pi t^2} \leftrightarrow e^{-\pi f^2}$$

$$u(t)$$

$$\text{sgn}(t) \leftrightarrow \frac{2}{j\omega}$$

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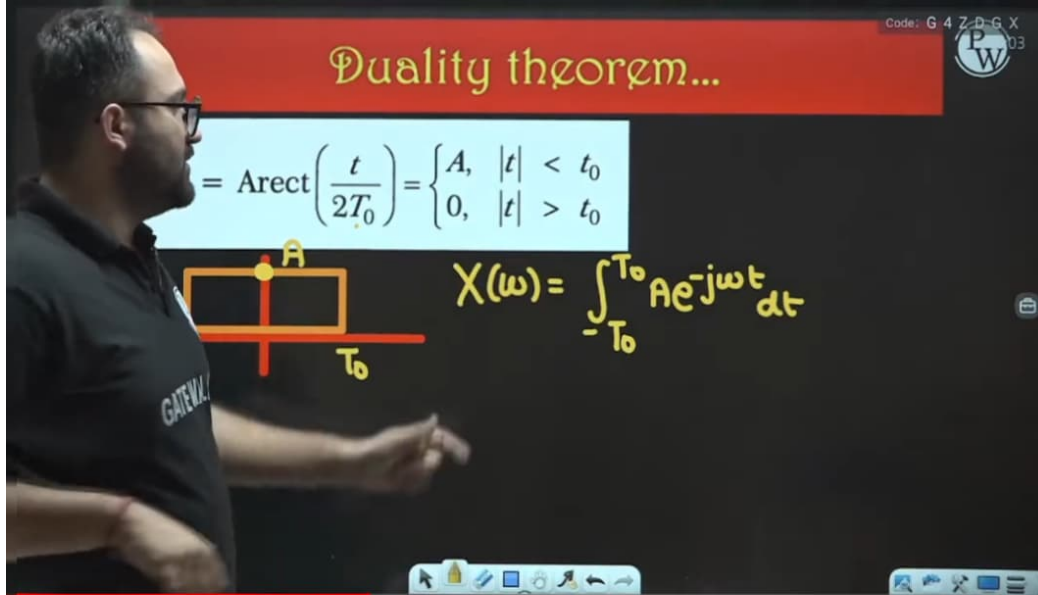


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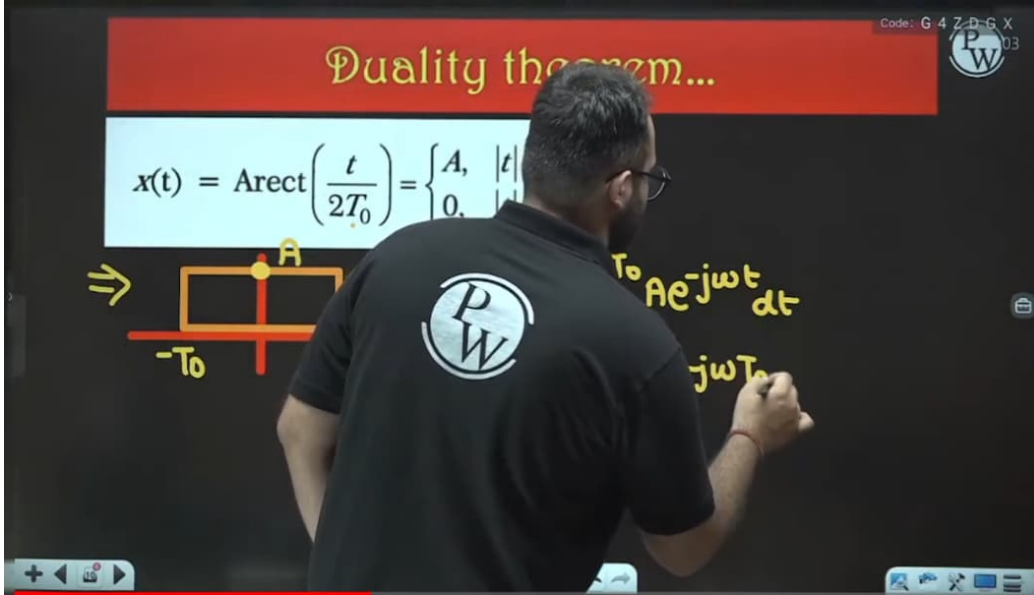
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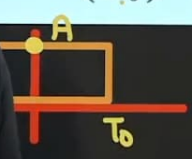


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Duality theorem...

$$x(t) = \text{Arect}\left(\frac{t}{2T_0}\right) = \begin{cases} A, & |t| < t_0 \\ 0, & |t| > t_0 \end{cases}$$


$$X(\omega) = \int_{-T_0}^{T_0} A e^{-j\omega t} dt$$

$$= \frac{A}{-j\omega} \left[e^{-j\omega T_0} - e^{+j\omega T_0} \right]$$

$$= \frac{2A}{\omega} \sin(\omega T_0)$$

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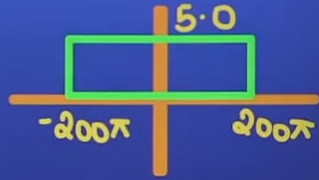


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$$\frac{5 \sin(200\pi t)}{\pi t} \longleftrightarrow 5 \text{rect}\left(\frac{\omega}{400\pi}\right)$$



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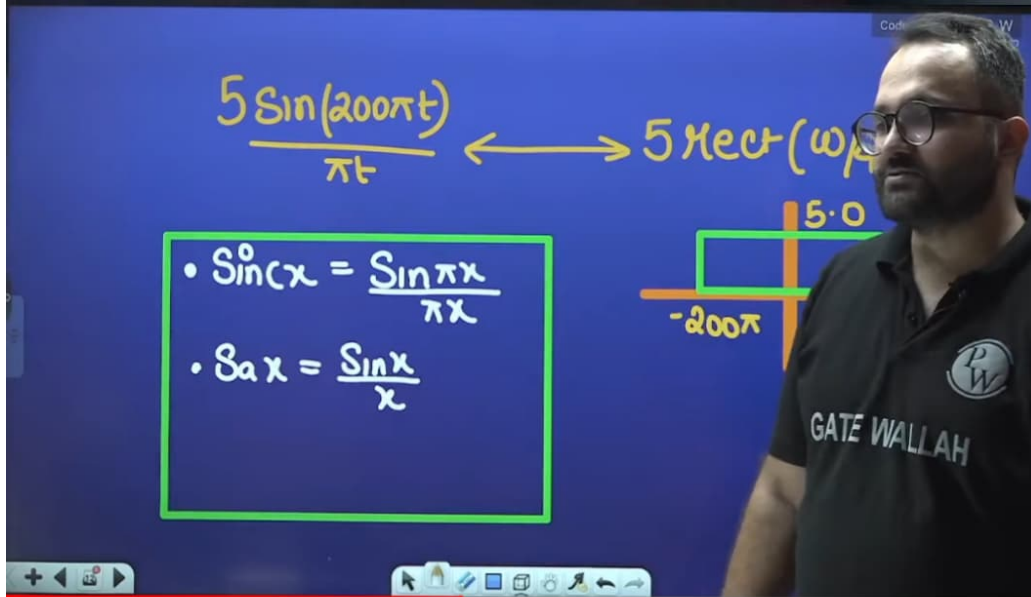


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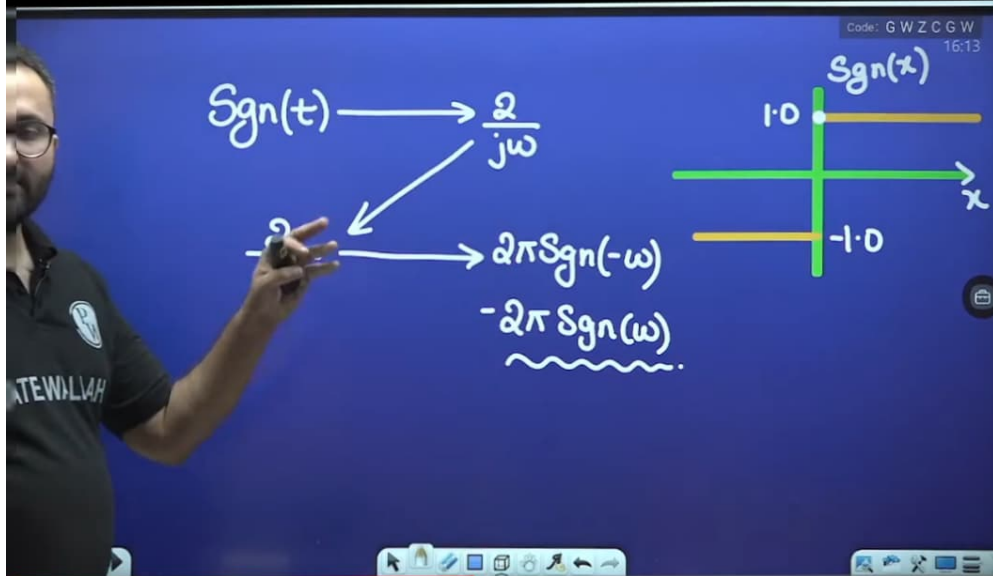
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@abhichandra7617 · 10 mo ago

t- 46:39 question me without solving likha sir and you are solving it



@logic727 · 9 mo ago

Great lecture 🔥



1



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@ayushbhowmick9752 · 1 yr ago

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$$\begin{aligned} & \cdot \frac{1}{jt} \leftrightarrow \text{Sgn} \\ & \cdot \frac{1}{a^2 + t^2} \leftrightarrow \\ & \cdot \frac{1}{jt - a} \\ & \cdot \frac{\mathcal{H} \sin(\omega_0 t)}{\pi t} \end{aligned}$$

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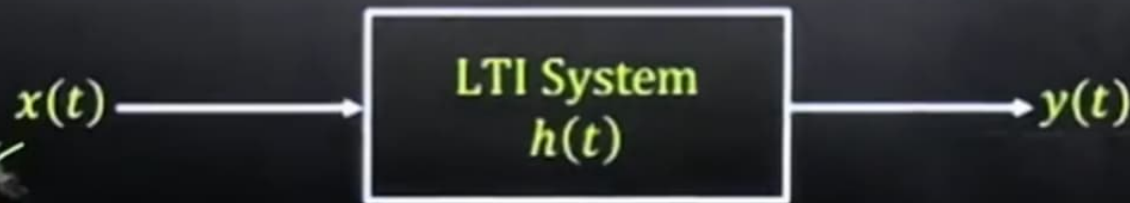


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Filter Concept (Band Pass Filter)

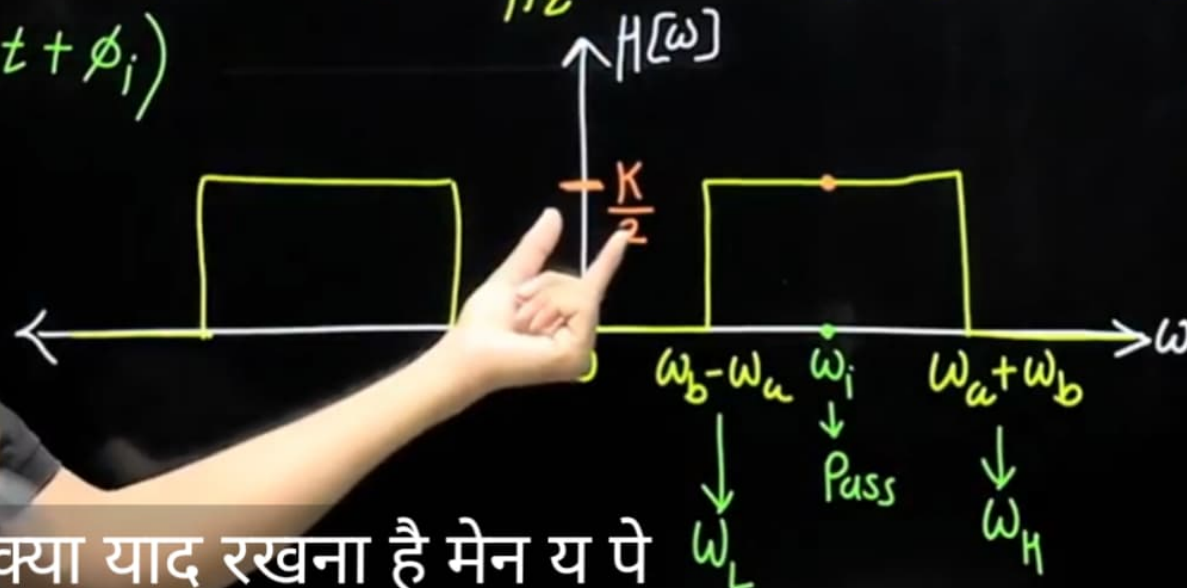


$$\omega_L = \omega_b - \omega_a$$

$$\omega_H = \omega_b + \omega_a$$

$$h(t) = K \cdot \frac{\sin \check{\omega}_a t}{\pi t} \cdot \cos \check{\omega}_b t ; \omega_b > \omega_a$$

$$x(\omega_i t + \phi_i)$$

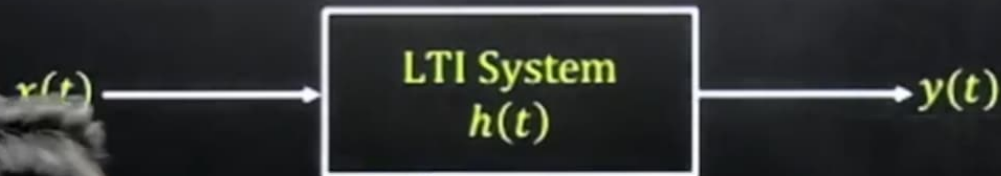


$$\textcircled{1} \omega_L \leq \omega_i \leq \omega_H$$

$$y(t) = \frac{K}{2} x(t)$$

याद रखिए इसमें क्या याद रखना है मेन य पे
के है तो जब मैग्नी मल्टीप्लाई

[Q] $x(t) = 2 \sin \pi t + 3 \cos 4\pi t - 6 \sin 10\pi t$. Find $y(t)$



$$h(t) = \frac{k \sin 6\pi t}{\pi t} \cos 8\pi t.$$

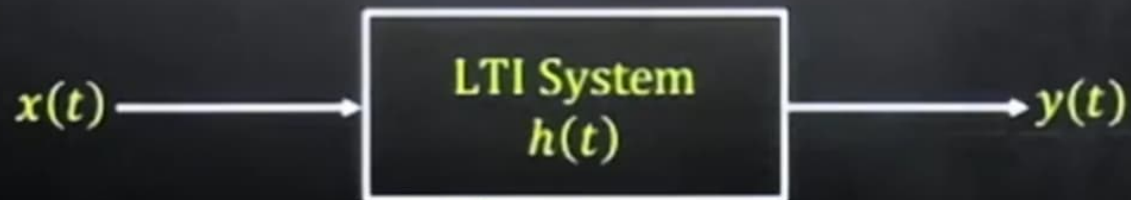
$$\omega_L = 8\pi - 6\pi = 2\pi$$

$$\omega_H = 8\pi + 6\pi = 14\pi$$

$$\therefore 2\pi \leq \omega_i \leq 14\pi.$$

गाइस यानी कि k इ 2 तो के बाटू यानी कि व
मल्टीप्लाई करोगे तो

[Q] $x(t) = 2 \sin \pi t + 3 \cos 4\pi t - 6 \sin 10\pi t$. Find $y(t)$



$$h(t) = \frac{K \sin 6\pi t}{\pi t} \cos 8\pi t.$$

$$\omega_L = 8\pi - 6\pi = 2\pi$$

$$\omega_H = 8\pi + 6\pi = 14\pi$$

$$Pass: 2\pi \leq \omega_i \leq 14\pi.$$

$$y(t) = 3 \cos 4\pi t - 6 \sin 10\pi t.$$

माइनस सि साइन 10 पाटी राइट दिस इ वेरी
वेरी वेरी मच इंपोर्टेंट

Fourier Transform of Periodic Signal

[Q] $x(t) = \sum_{n=-\infty}^{n=\infty} A\delta(t - nT)$. Find FT of $x(t)$

$$\Rightarrow x_p(t) = \sum_{n=-\infty}^{n=\infty} C_n e^{jn\omega_0 t} \Rightarrow C_n = \frac{1}{T} \int_{-\infty}^{+\infty} x_p(t) \cdot e^{-jn\omega_0 t} \cdot dt$$

$$\Rightarrow X_p[\omega] = \sum_{n=-\infty}^{n=\infty} C_n \cdot 2\pi \delta[\omega - n\omega_0]$$

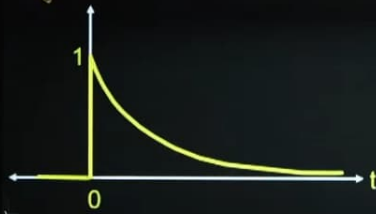
$$1 \xleftarrow{FT} 2\pi \delta[\omega]$$

$$e^{jn\omega_0 t} \xleftarrow{FT} 2\pi \delta[\omega - n\omega_0]$$

पीरियोडिक सिग्नल कोई भी पीरियोडिक सिग्नल
दिया जाए तो पहले C फैडुगुरू



Fourier Transform of Some Standard Signals ✓



$$x(t) = e^{-at}u(t) \xrightarrow{\text{FT}} X[\omega] = \frac{1}{a + j\omega}$$

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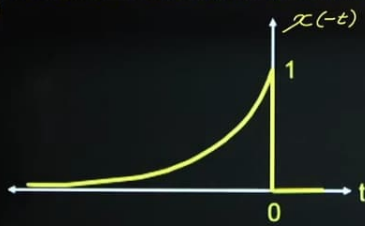
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Fourier Transform of Some Standard Signals



$$x(t) = e^{at}u(-t) \xrightarrow{\text{FT}} X[\omega] = \frac{1}{a - j\omega}$$

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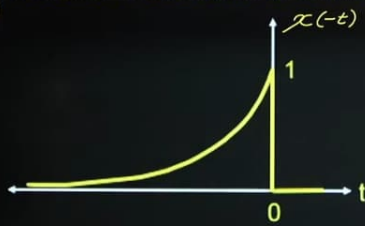
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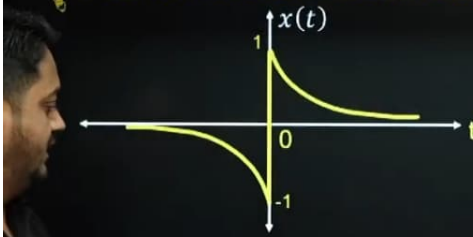
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Fourier Transform of Some Standard Signals



$$x(t) = e^{-at}u(-t) \xrightarrow{\text{FT}} X[\omega] = \frac{-2j\omega}{a^2 + \omega^2}$$

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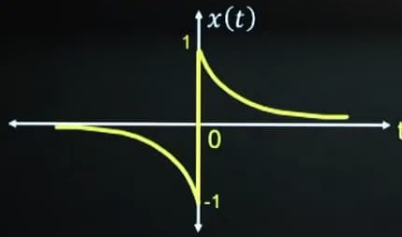
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Fourier Transform of Some Standard Signals



$$x(t) = e^{-at}u(t) - e^{at}u(-t) \xrightarrow{\text{FT}} X[\omega] = \frac{-2j\omega}{a^2 + \omega^2}$$



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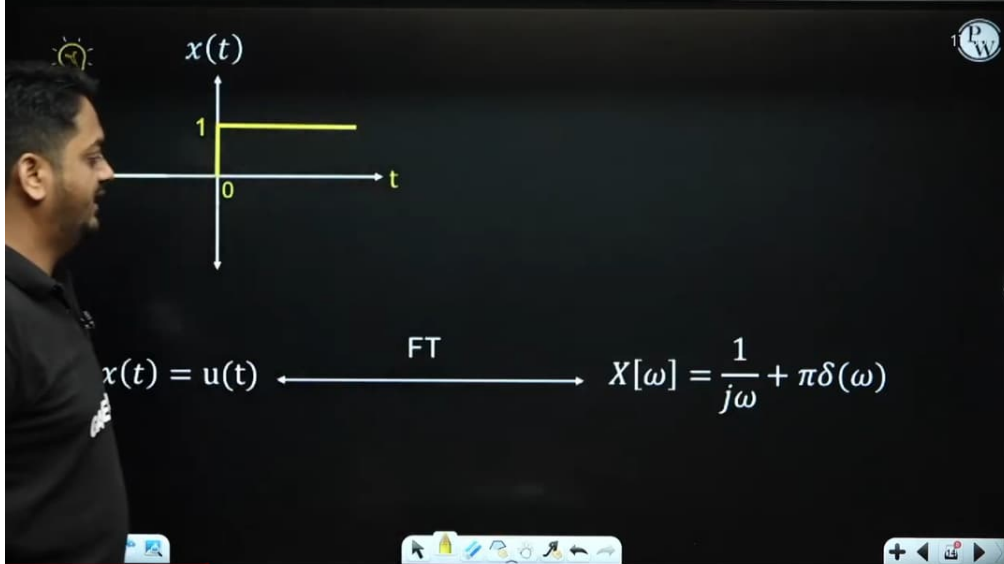
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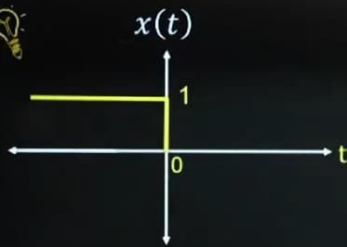
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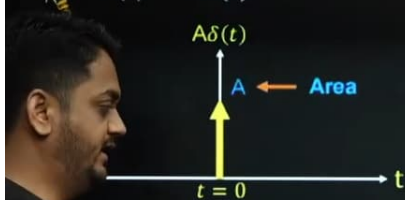
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$$x(t) = A\delta(t)$$



$$x(t) = A\delta(t)$$

FT

$$X[\omega] = A$$

$$A$$

FT

$$X[\omega] = A2\pi\delta[\omega]$$

$$X[f] = A\delta[f]$$

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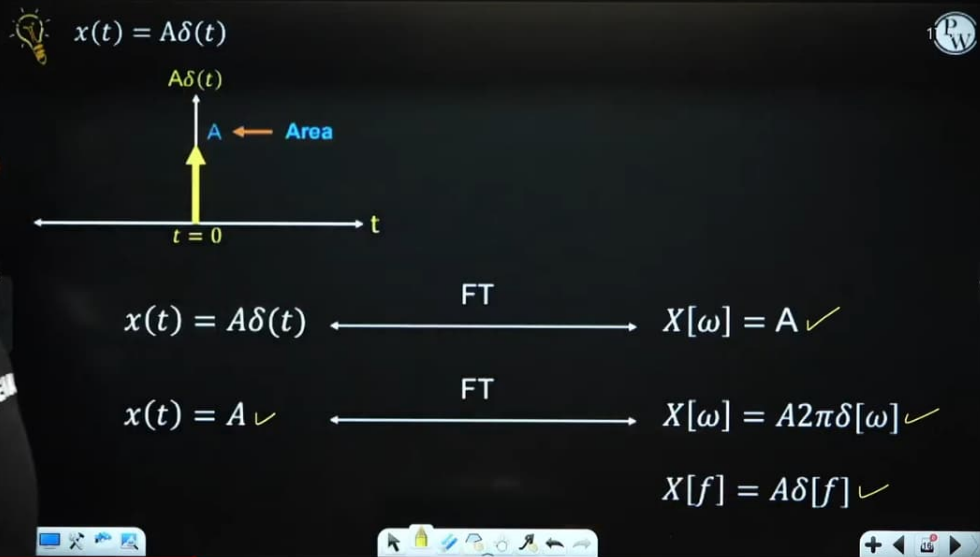
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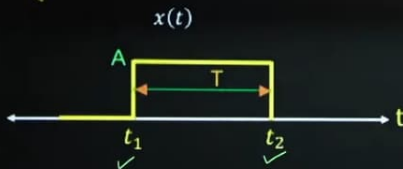


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💡 Very much Important Short Cut ★★



$$X[\omega] = e^{-j\omega\left(\frac{t_1+t_2}{2}\right)} AT \text{sinc}\left(\omega \frac{T}{2}\right)$$

$$X[f] = e^{-j2\pi f\left(\frac{t_1+t_2}{2}\right)} AT \text{sinc}(fT)$$

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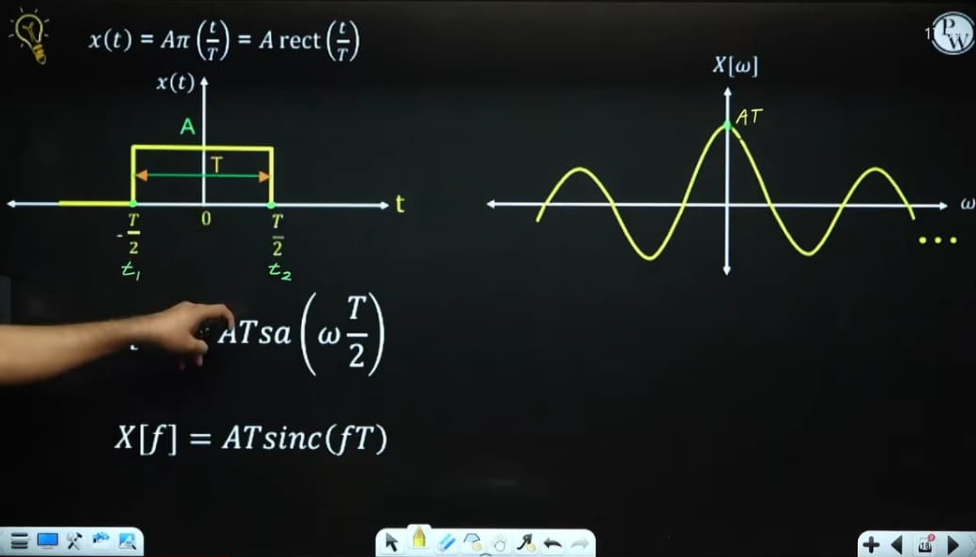
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$$x(t) = \cos \omega_0 t \xrightarrow{\text{FT}} X[\omega] = \frac{2\pi\delta[\omega - \omega_0] + 2\pi\delta[\omega + \omega_0]}{2}$$

$$X[f] = \frac{\delta[f - f_0] + \delta[f + f_0]}{2}$$

$$x(t) = \sin \omega_0 t \xrightarrow{\text{FT}} X[\omega] = \frac{2\pi\delta[\omega - \omega_0] - 2\pi\delta[\omega + \omega_0]}{2j}$$

$$X[f] = \frac{\delta[f - f_0] - \delta[f + f_0]}{2j}$$

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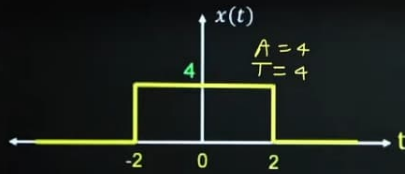
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[Q] Find FT of $x(t)$.

$$X[\omega] = e^{-j\omega\left(\frac{0}{2}\right)} \cdot 16 \cdot \text{sinc}\left(\omega \cdot \frac{4}{2}\right) = 16 \text{sinc}(2\omega)$$

$$X[f] = 16 \text{sinc}(4f)$$

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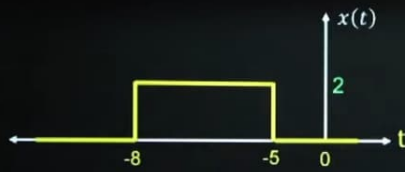


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[Q] Find FT of $x(t)$.



$$X(\omega) = e^{-j\omega\left(-\frac{13}{2}\right)} \cdot 6 \cdot \text{sinc}\left(\omega \cdot \frac{3}{2}\right) = e^{j\frac{13\omega}{2}} \cdot \text{sinc}\left(\frac{3\omega}{2}\right) \cdot 6$$

$$X(f) = e^{j13\pi f} \cdot \text{sinc}(3f) \cdot 6$$

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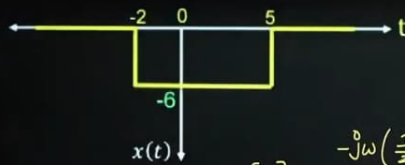


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[Q] Find FT of $x(t)$.



$$X(\omega) = e^{-j\omega\left(\frac{3}{2}\right)} \cdot (-42) \cdot \text{sinc}\left(\omega \cdot \frac{7}{2}\right)$$

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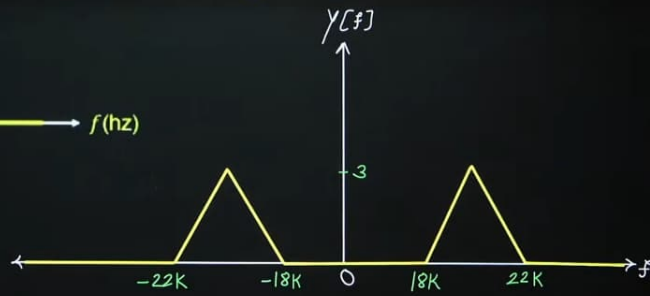
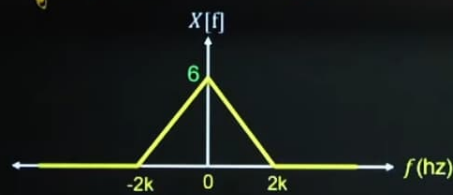
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[Q] If $x(t) \xrightarrow{FT} X[f]$. Find FT of $y(t) = x(t) \cos 4\pi 10^4 t$.



$$f_0 = 20 \text{ KHz}$$



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Properties of Fourier Transform



$$x(t) \xleftrightarrow{FT} X[\omega]$$

Duality Property ★ ★ ★

$$x(t) \xleftrightarrow{FT} X[\omega] \iff X[f] \xleftrightarrow{FT} x[t]$$

$$X[t] \xleftrightarrow{FT} 2\pi x[-\omega]$$

$$X[t] \xleftrightarrow{FT} x[-f]$$

$$e^{-at} u(t) \xleftrightarrow{FT} \frac{1}{a + j\omega}$$

$$\frac{1}{a + jt} \xleftrightarrow{FT} 2\pi e^{a\omega} u[-\omega]$$

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$x(t) = K \frac{\sin at}{\pi t}$

Area of $K \frac{\sin at}{\pi t} = K$

Energy of $K \frac{\sin at}{\pi t} = K^2 \frac{a}{\pi}$

स्केलिंग हुआ इसलिए के स्क के स्क्र क्यों आया एंप्लीट्यूड स्केलिंग गा ओके

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